SummaRace v3

Read it · Race it · Sum it up!

Handover Package

Product, technical, design and data documentation for the offline Grade 4 reading game, with the asset credits and the full answer key.



Software version	1.3.6 (versionCode 17), Android app `com.summarace.app`
Package version	1.0
Date	10 November 2026
Prepared for	[Client / research team name]
Prepared by	[Author full name] · github.com/orbitsdev

BEFORE YOU START

Document control

Document information

Title	SummaRace v3 Handover Package
Describes	The `main` branch of github.com/orbitsdev/summaracev3 at version 1.3.6 (versionCode 17). `main` has not changed since 5 November 2026.
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Revision history

Version	Date	Change
0.1	5 Nov 2026	First set of separate documents: PRD, TRD, App Flow, Design Brief, Data Schema, Implementation Plan, Assets.
0.2	6 Nov 2026	Compiled into one file; answer key added from `stories.js`.
0.3	10 Nov 2026	System Design chapter added; all chapters re-checked against `main`.
1.0	10 Nov 2026	Handover package: cover, document control, executive summary, numbered figures and tables, glossary, references and appendices.

Sign-off

Role	Name	Signature	Date
Prepared by (developer)			
Received by (client)			
Reviewed by (adviser, if any)			

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> The answer key tables in Appendix B are not numbered; each story has its own section there.

OVERVIEW

Executive summary



SummaRace is an offline reading game for Grade 4 learners. For each story the learner **reads** it page by page, **races** to pick its story parts in a 3D runner, puts the parts **in order**, and then **sums it up** in 2–3 sentences, guided by Teacher Mira.

30	5	4	0
stories in 10 sessions	worlds, 6 levels each	stages per level	servers or accounts

What was built

Version 1.3.6 is a web game (HTML, CSS, JavaScript and three.js) inside a small Android app. It runs fully offline on Android 7.0 or newer. Learner progress is saved on the phone, and the teacher exports one CSV row per finished level for the research team. The summary is checked offline by rules (no AI model, no internet), and Teacher Mira speaks with 320 pre-recorded voice clips.

What this package contains

- **Chapters 1–7** describe the product (PRD), the technology (TRD), the screen flow, the system design, the look and feel, the data and the order in which it was built.
- **Chapter 8** lists what a laptop, PC or phone needs to run the game, with setup steps. **Chapter 9** lists what is handed over and the items still open.
- **Appendix A** lists every asset the game ships, with its source, author and license. **Appendix B** is the answer key for all 30 stories.

Status at handover

The game is feature-complete. Before it goes to learners, three things need attention (details in Chapter 9):

1. Release 1.3.7 is signed with a new key, so every phone needs its records exported and the old app uninstalled first, and the release still needs a full play-through on a real phone.
2. The music and all UI images (including Teacher Mira) have no recorded source, and some required credits (wordfreq, font licenses) are not yet in the game.
3. The research team should confirm permission and credit lines for the stories, several of which are published children's books.

CHAPTER 1

Product Requirements

SummaRace v3 · version 1.3.6 · Android app `com.summarace.app`

1.1 Overview

This document describes what SummaRace does, feature by feature, and how each feature is expected to behave. It covers the game as built in version 1.3.6, not planned features.

SummaRace is a reading game for Grade 4 learners. For each story the learner **reads** it page by page, **races** through its story parts in a 3D runner, puts the parts **in order**, and then **sums it up** in 2–3 sentences. The tagline on the title screen is "Read it · Race it · Sum it up!". Teacher Mira, a cartoon teacher, guides the learner with spoken lines the whole way.

1.2 Product goal

Help Grade 4 learners practise reading comprehension and summarising with the **Somebody – Wanted – But – So – Then** (SWBST) method, in a game they want to keep playing, and give the teacher a record of every finished level that they can open in a spreadsheet.

The game has to work on low-cost Android phones in a classroom with no internet and no accounts.

1.3 Users

Learner

A Grade 4 pupil. Types their name and section, plays the stories in order, earns stars and coins.

Teacher

Hands out the phones, reads the records screen, exports the CSV, can erase or bring back records.

Research team (client)

Wrote the 30 stories, questions and SWBST answers. Uses the CSV results for the study.

1.4 Content scope

- **30 stories** in **10 sessions**. Each session has one easy, one average and one hard story (levels 1–30).
- Each story has **5 pages**, **5 page questions** (one per page), **1 main idea question**, and 5 **SWBST story parts** with right and wrong answers.
- The 30 levels sit in **5 worlds** of 6 levels each: Sunny Meadow, River Bend, Story Town, Palm Beach and Summit Peak.
- The pages, page questions, SWBST answers and main idea come word for word from the research team. The wrong answer choices, race wording and Teacher Mira's example summaries were written for the game.

1.5 Features

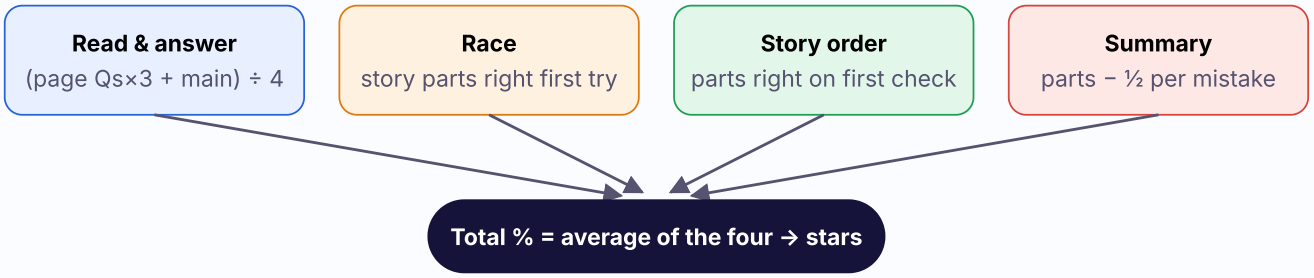
Table 1.1 Features

Feature	Description	Expected behaviour
Title screen	Animated title with Teacher Mira, a PLAY button and Credits.	PLAY opens "Who is playing?". Credits opens the credits popup.
Learner profiles	Name (required, max 40 characters) and section (optional, max 30).	A new name and section makes a new profile on this device (matching ignores capital letters). Returning learners tap their name under "Or continue as". Progress is saved per profile.
World map	A scrolling path of 30 level nodes with a session bar (sessions 1–10) and a "go to my level" button.	Levels unlock one by one. Tapping a locked level shows a card that sends the learner to their open level. Stars and best score show on finished levels.
Read	One page at a time with a world picture, page dots and a Listen button.	Mira says a short tip and then reads the page aloud with a recorded voice. "Answer the question" goes to that page's question.
Page questions	3 answer choices in a new random order every time.	A wrong answer is never revealed. After a wrong answer Mira offers "Read the story" or "Try again". After 2 wrong tries the story opens with the right part highlighted. Only a right first try counts for the score.
Main idea	After the last page, 4 choices: which one tells what the whole story is about.	Same retry rules as page questions. Scored right or wrong on the first try.
Race	A 3D endless runner: the learner's runner (Max or Mia) escapes a police officer and dog.	Each SWBST part becomes a question at the top with answers A, B and C shown in the 3 lanes. The learner swipes into the lane of the right answer and runs through its gate. Right = boost and the patrol falls back. Wrong = the patrol comes closer and the question comes back later. Trains and barriers must be dodged (swipe left/right, up to jump, down to roll). If the patrol catches the runner, the race restarts.
Story order	Tap the 5 story parts into the Somebody, Wanted, But, So, Then slots.	Right parts turn green and stay. After the first wrong check a 30-second clock starts (it pauses while the story or Settings is open). When time runs out Mira shows the right order and the level goes on. Score = parts right on the first check.
Summary	The learner writes 2–3 sentences in English, in their own words.	An offline checker gives feedback lines (see TRD). It passes when the summary names the main character, has most story parts and the main idea, and has no clear mistakes. After 3 tries Mira shows an example. After 4 tries the learner can choose "Finish, my teacher will check".
Results	Stars (1–3), a title, the score for each stage and coins earned.	Total ≥ 85% = 3 stars "AMAZING!", ≥ 65% = 2 stars "GREAT JOB!", otherwise 1 star "GOOD TRY!". Buttons: Map, Replay, Next. The next level unlocks.
Teacher records	A table of every finished level on this device.	Copy CSV, Download CSV (saved to `Download/SummaRace/` on Android), Share CSV (Android share sheet). "Erase all records" needs the word ERASE typed in; the last erase can be undone with "Bring back erased records".
Settings	Teacher voice, sound effects and music switches; Change player; Teacher records; Quit this story; Exit game.	Switches are saved on the device. Quit is offered during a level, Exit game only inside the Android app.
Teacher Mira	A guide docked at the bottom of every screen except the race and records.	Speaks recorded lines and shows a speech bubble. The bubble folds away after 7 seconds, or after 2 seconds if it covers a button the learner needs. Tap her to hear the line again.
Android Back button	Handled by the game, not the system.	Closes a popup first. In the race, the first Back pauses. During a level it opens the Quit popup instead of dropping progress. On the map it goes back to the title screen.

1.6 Scoring

The level total is the average of four stage scores:

Figure 1.1 Total score is the average of four parts



Note. Questions, race and story order count the first try only, so retries help the learner finish but do not raise the score. The summary counts the try that passed, minus half a part for each wrong detail. "Finish, my teacher will check" scores the summary as 0 and leaves it for the teacher.

Coins collected in the race are added to the learner's coin total. A replay keeps the best stars and best score for that level.

1.7 Constraints

- **Fully offline.** No server, no login, no analytics, no ads. Everything ships inside the APK.
- Android 7.0 or newer (minSdk 24), portrait only, phones with about 2 GB of RAM.
- Records live only on the device they were made on. The teacher must collect the CSV from each phone.
- The summary checker must run offline, so it uses word lists and rules, not an AI model.

1.8 Out of scope

- Syncing records between devices or to the cloud.
- Teacher accounts, passwords or a class dashboard.
- Languages other than English (the summary must be written in English).
- iOS builds (the web game would run in a browser, but no iOS app is made).

1.9 Open items

Table 1.2 Open items in the product scope

Item	Status
Two stories are both titled "In Grandfather's Day" (levels 15 and 18).	Needs the client
Profiles are matched by name and section, so two learners with the same name in the same section on one phone share a profile.	Open
Some newer Mira lines have no recorded clip and fall back to the phone's voice. The clip build script was lost.	Open
PR #4: race music plays faster as story parts are collected.	Waiting for a decision
PR #5: Story Stop voice lines. No longer needed since the race was changed to lane answers.	Can be closed

CHAPTER 2

Technical Requirements

SummaRace v3 · version 1.3.6 (versionCode 17)

2.1 Overview

This document lists the tools, technologies and structure used to build SummaRace v3, so anyone who picks up the code knows how it fits together and what it must keep doing.

SummaRace v3 is a web game (HTML, CSS and plain JavaScript, with three.js for the 3D race) that runs inside a small Android app. The Android app is a full-screen WebView plus a JavaScript bridge for things a web page cannot do on its own: the phone's voice, the clipboard, sharing, saving files and closing the app.

2.2 Technical goals

- **Works with no internet.** Every file is inside the APK. Nothing is fetched from the network.
- **Runs on low-cost phones.** Android 7.0+ (API 24), about 2 GB RAM. The 3D models are not loaded at start-up (they preload in the background after the first tap), and GPU memory is freed after each obstacle.
- **Easy to change.** No build step for the game. Stories are one data file. Opening the folder with a local web server runs the game in a desktop browser.
- **Safe for learners.** No accounts, no tracking, no ads, no outside links during play.

2.3 Technology stack

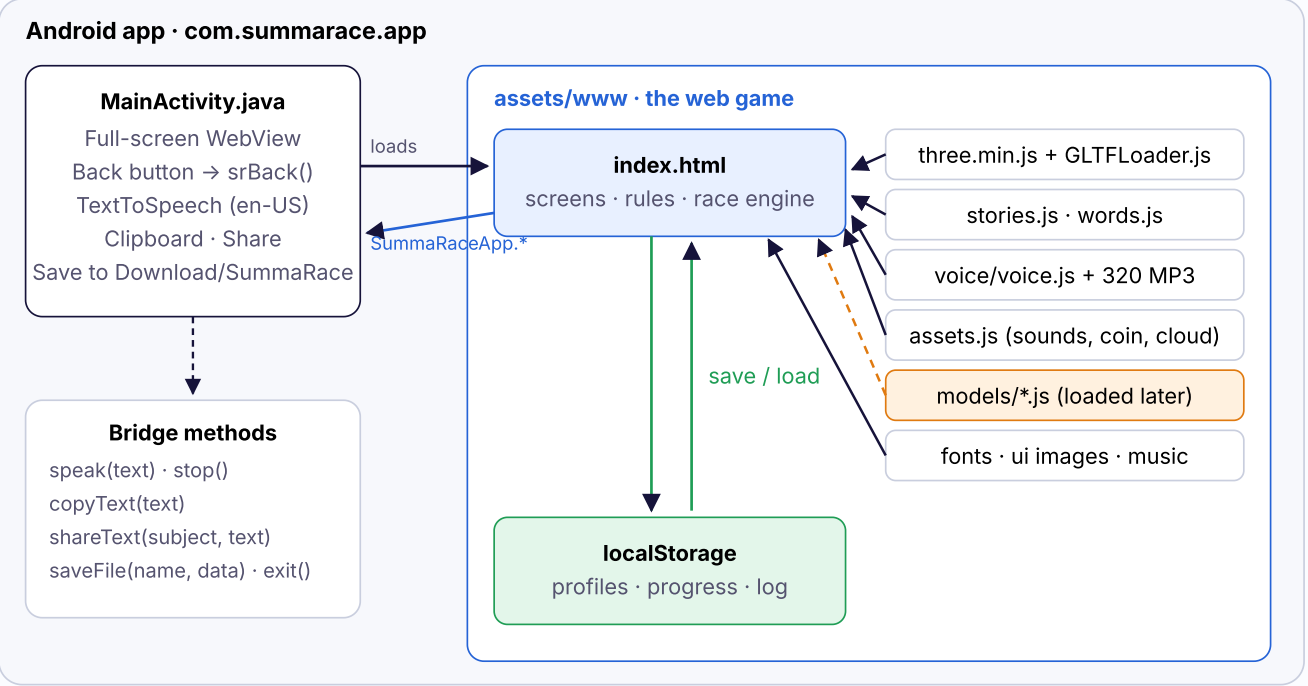
Table 2.1 Technology stack

Area	Tool / technology	Purpose
Game UI	HTML5, CSS, plain JavaScript (ES2018), one `index.html`	Every screen, popup, Teacher Mira, quiz, order and summary logic. No framework.
3D race	three.js r128 (`three.min.js`) + `GLTFLoader.js`	The endless-runner race: toon shading, animated characters, trains, scenery.
3D models	GLB files stored as base64 text in `models/*.js`	Loads from inside the app with no file-access problems. Compressed with glTF-Transform 4.1.1.
Sound effects	Web Audio API, clips in `assets.js` as data URLs	Click, coin, jump, bump, good, bad, win and so on.
Music	HTML `Audio` elements, looping MP3	`menu.mp3` on all screens, `race.mp3` in the race. Ducks under Mira's voice.
Voice	320 pre-recorded MP3 clips (Kokoro TTS) + `voice/voice.js` index	Mira's lines and all 150 story pages read aloud. Falls back to Android TextToSpeech (en-US) or the browser's speech when a line has no clip.
Story data	`stories.js` (global `window.STORIES`)	30 stories with pages, questions, SWBST parts, main idea and model summary.
Word list	`words.js` (global `window.SR_WORDS` , 24,887 words)	Checks that a summary is written in real English words.
Storage	Browser `localStorage` (inside the WebView)	Profiles, progress, settings and the attempt log. See the Data Schema.
Android host	Java, one `MainActivity` , Android WebView	Full-screen portrait WebView, Back button, bridge `window.SummaRaceApp` .

Area	Tool / technology	Purpose
Build	Gradle, Android Gradle Plugin 8.7.3, compileSdk / targetSdk 35, minSdk 24	Builds the APK. MP3 and WOFF2 files are stored uncompressed so the WebView can read them.
Version control	Git + GitHub (`orbitsdev/summaracev3` , public)	Source, pull requests and history.

2.4 Architecture

Figure 2.1 Architecture diagram



Note. The dashed orange arrow is a lazy load: the 3D model files are not loaded at start-up; they preload in the background 1.5 s after the first tap. No arrow leaves the phone: there is no server.

Script load order

`three.min.js` → `GLTFLoader.js` → `assets.js` → `stories.js` → `words.js` → `voice/voice.js` → the main script inside `index.html` . The main script is wrapped in one function (an IIFE), so its internals are not global. Each data file sets one global: `SR_ASSETS` , `STORIES` , `SR_WORDS` , `SR_VOICE` , `SR_MODELS` .

Main code modules inside index.html

Table 2.2 Main code modules inside index.html

Module	What it does
Storage	`loadDb()` , `persist()` , preferences. One JSON object under `summarace.v1` .
Teacher Mira	`say()` shows a line and speaks it. Picks a recorded clip by normalised text (`vnorm`), maps lines with names or numbers to a recorded version, avoids repeating the same line, and folds her bubble when it covers a button.
Audio	Sound effects on Web Audio, music on looping `Audio` elements, voice with two priorities (story reading is never cut off by a short line).

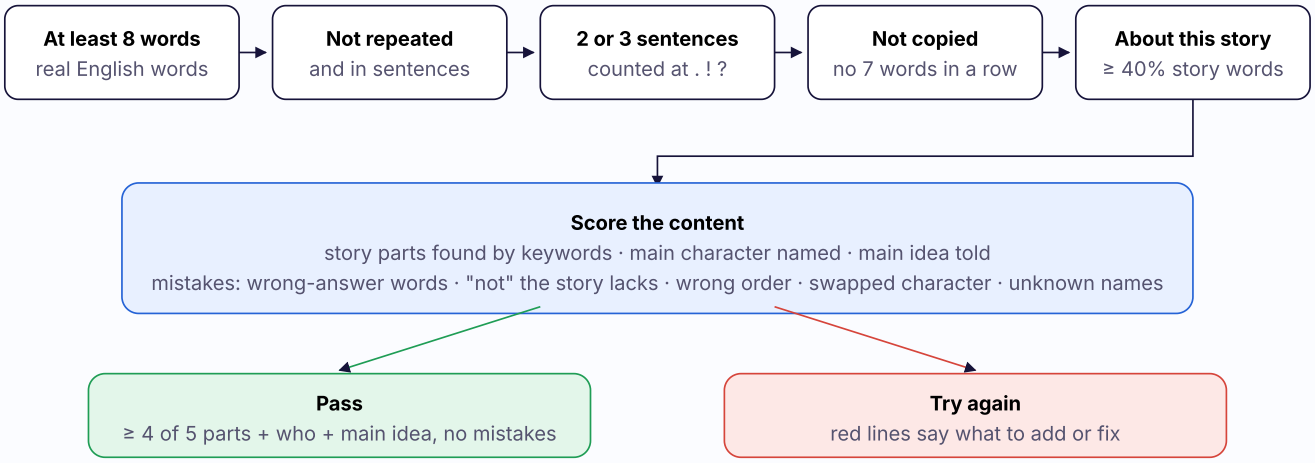
Module	What it does
Screens	`show(id)` switches sections: learner, home, read, quiz, order, race, summary, result, records.
Level flow	`startLevel` → `renderPage` / `askPage` → `askMain` → `startRace` → `startOrder` → `startSummary` → `finishLevel`.
Summary checker	Offline rules, see section 6.
Race	A self-contained module (`Race.prepare` , `Race.stop`) with the three.js scene, worlds, characters, obstacles, gates, patrol and end scenes.
Records	Table, CSV export (copy, download, share), erase with undo.

2.5 Race engine

- Three lanes at x = −2.2, 0 and 2.2. The runner (Max or Mia, chosen on the start card) is chased by an officer and a dog.
- A **gap** meter (0–100) is the distance to the patrol. It starts at 60 and shrinks slowly while running, but not while the learner is reading a question. A right answer adds 24. A wrong answer, a train bump or a barrier hit takes 10–14 away, but these never push the gap below 8: only the slow drain can end the race. At 0 the runner is caught and the race restarts.
- Each of the 5 SWBST parts is asked in turn. The question and the 3 answers show at the top. The game waits a reading time of 0.35 s per word (4–9 s, 25% less when a missed question comes back), then sends an answer gate down the track. The lane the runner is in at the gate is the answer.
- Each level has its own seeded course (`levelPlan(id)`): speed, obstacle rows and spacing grow with the level number, so level 30 is harder than level 1 but each level plays the same every time.
- Controls: swipe left/right to change lane, up to jump, down to roll. Tapping the left or right third of the screen also changes lane. Keyboard arrows (or A and D) and Space work in a desktop browser.
- Race score for the level = share of the 5 parts answered right on the first try, kept across "Try the race again".

2.6 Summary checker (offline)

Figure 2.2 Summary checker steps



Note. A failed early check (top row) does not count as a try. From the 3rd scored try, mistakes no longer block a pass, and Mira shows her example. From the 4th, "Finish, my teacher will check" appears.

The checker is rules-based by design, because it has to work offline on low-cost phones. It can miss things a teacher would notice, which is why the summary text and its status go into the CSV for the teacher to read.

2.7 Performance and device rules

- Start-up was measured at about 0.75 s with the CPU slowed 6× in a desktop browser. The 3D models (4.5 MB) are not part of start-up.
- Obstacles and effects free their GPU geometry and materials when removed; shared models and textures are kept.
- Portrait only. The layout uses safe-area insets for phones with a notch.
- Hardware acceleration is on for the WebView.

2.8 Build, run and release

```
Run in a browser:
  cd app/src/main/assets/www
  python -m http.server 8000      # then open http://localhost:8000

Build the APK:
  open the repo in Android Studio and build, or run: gradle assembleDebug
  the APK is written to app/build/outputs/apk/
```

- Raise `versionCode` and `versionName` in `app/build.gradle` for every release. The next free versionCode is 18.
- Release APKs are signed with the project's current signing key, which is kept off GitHub on the owner's PC. The original 1.2.0 key was lost, so phones with that old test build must uninstall it once.
- Never commit the keystore, `*.jks`, `*.keystore` or `local.properties`. They are in `.gitignore`.

2.9 Testing

- Day-to-day testing is in a desktop browser at `localhost:8000`, with the phone size set in the browser's device mode.
- Automated checks use Playwright with headless Chromium. Because the main code is private inside one function, tests reach it by injecting a small helper before the script ends.
- An APK is built for a phone test only when a change is ready. Android-only features (voice fallback, Download folder on Android 7–9, Back button) must be checked on a real phone.

CHAPTER 3

APP FLOW

SummaRace v3 · version 1.3.6

3.1 Overview

This document shows what happens when someone taps, and which screen comes next. It maps the main journeys through the game, from opening the app to finishing a level and exporting records.

3.2 Key user journeys

First play

Open app → PLAY → type name and section → World map → Level 1 → read, race, order, summarise → Results → Next.

Returning learner

PLAY → tap the learner's name under "Or continue as" → the map scrolls to their open level → Play.

Replay for more stars

Map → tap a finished level → Play again. Best stars and best score are kept.

Teacher collects results

PLAY → Teacher records (or Settings → Teacher records) → Copy, Download or Share CSV.

Stop mid-level

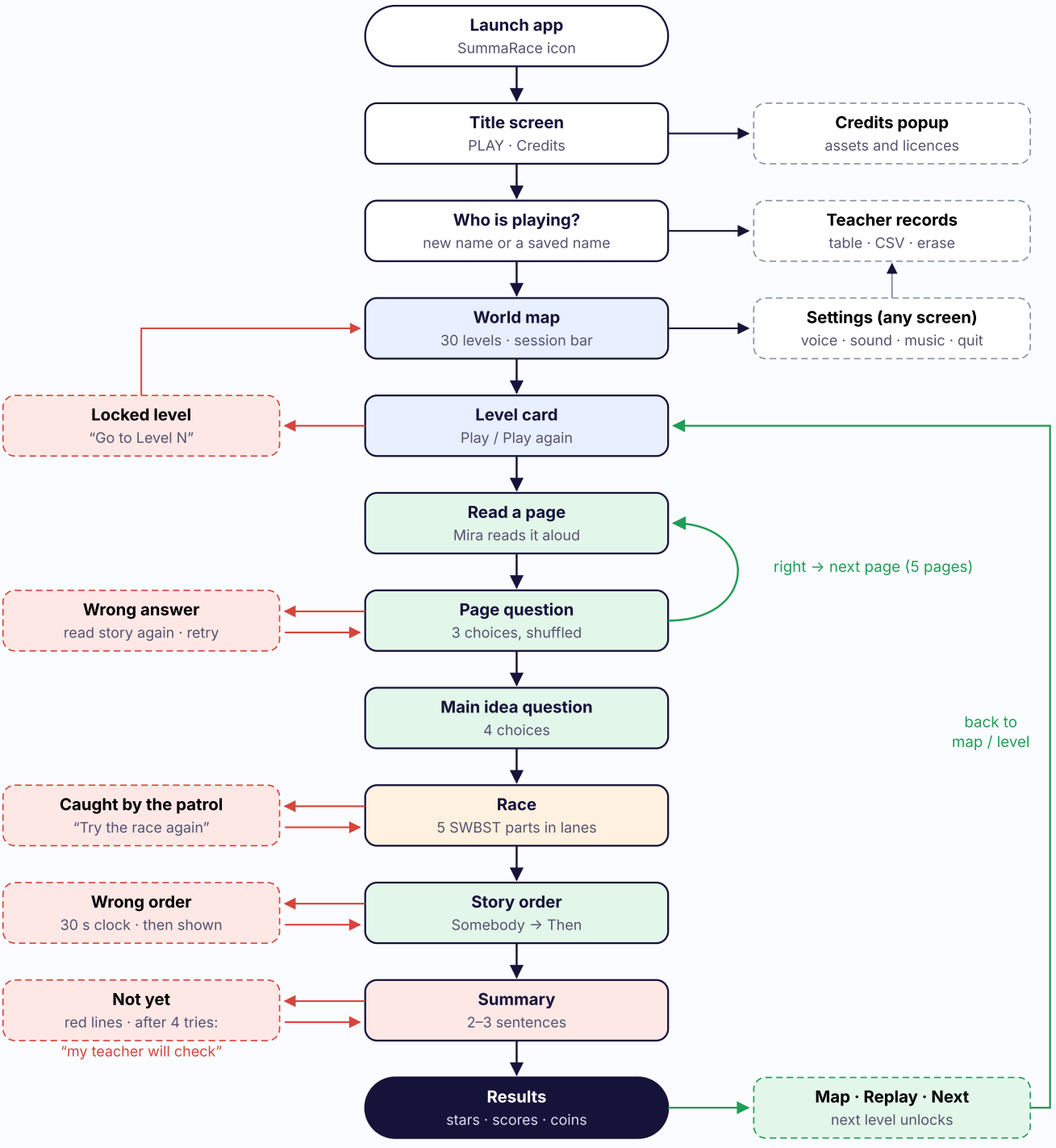
Settings or the Back button → Quit this story → World map. The unfinished level is not recorded.

Change player

Settings → Change player (or tap the name chip) → "Who is playing?".

3.3 App flow diagram

Figure 3.1 Main app flow from launch to results



Note. Solid boxes are the main path. Dashed boxes are side screens and retry loops. Green arrows move forward, red arrows are a retry or a block.

3.4 Inside one level

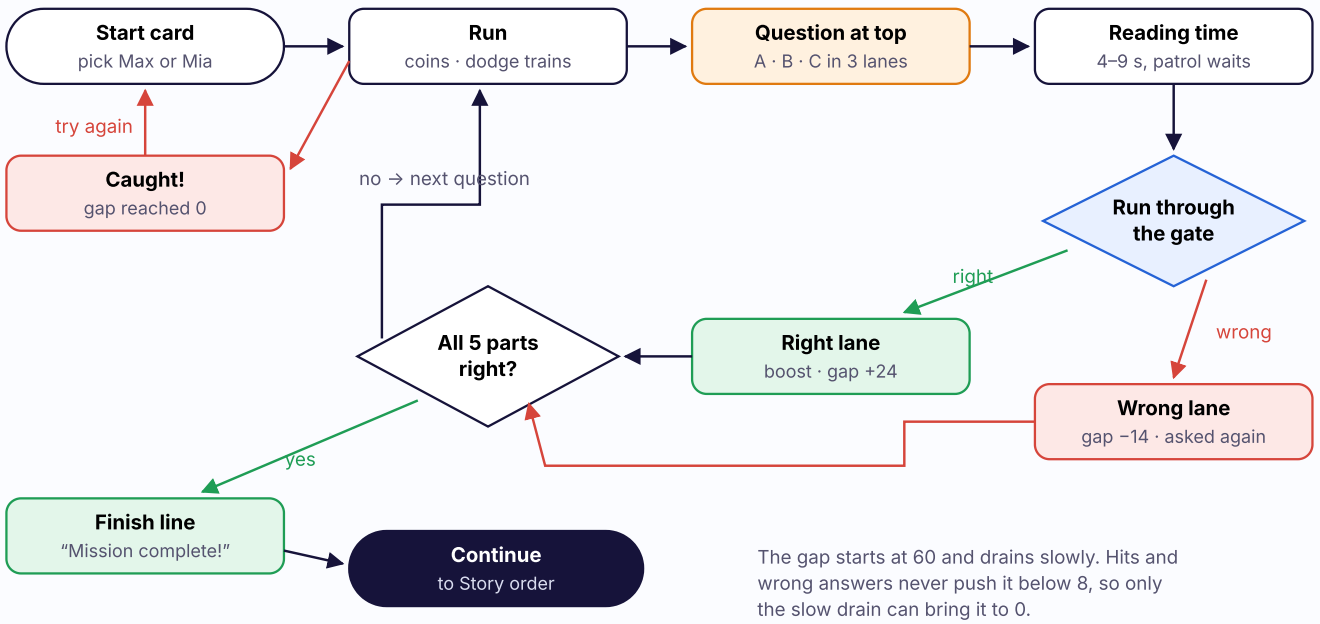
A level always runs in the same four steps, shown at the top of each screen as a step bar: **Read & answer → Race → Story order → Summary**.

Table 3.1 Inside one level

Step	Screen	What the learner does	What happens next
1	Read	Reads page 1 of 5 (Mira reads it aloud too). Taps "Answer the question".	That page's question.
1	Page question	Picks 1 of 3 answers.	Right: "Next page" (or "Next: the main idea" after page 5). Wrong: Mira offers to show the story; after 2 wrong, the story opens with the part highlighted in yellow.
1	Main idea	Picks 1 of 4 answers.	Right: "Next: the race!".
2	Race	Start card → Start running. Answers 5 story-part questions by lane.	Mission complete → Continue.
3	Story order	Taps the 5 parts into Somebody, Wanted, But, So, Then. Taps "Check order".	All right: goes on by itself. Some wrong: green parts stay, 30-second clock, try again.
4	Summary	Writes 2–3 sentences. Taps "Check my summary".	Pass: "See my results". Not yet: feedback lines, try again.
—	Results	Sees stars and scores.	Map, Replay, or Next (hidden after level 30).

3.5 Race flow

Figure 3.2 Race loop



Note. The race is the only place a learner can "lose", and losing only restarts the race. The level score keeps first-try answers from every attempt.

3.6 Unlock rules

- A new profile starts with Level 1 open.
- Finishing level *n* (any score) opens level *n* + 1. Levels open strictly one by one.
- The session bar on the map groups levels in threes (session 1 = levels 1–3). A session chip is locked until its first level is open.
- Level *n* belongs to world ⌊(n − 1) ÷ 6⌋: levels 1–6 Sunny Meadow, 7–12 River Bend, 13–18 Story Town, 19–24 Palm Beach, 25–30 Summit Peak. The world sets the race scenery and the screen backdrop.

3.7 Back button

Table 3.2 Back button

Where	Android Back does
Any popup open	Closes the popup.
Race, running	Pauses the race.
Race, any card showing	Opens the Quit popup.
Read, question, order, summary, results	Opens the Quit popup (Quit this story / Exit game), so a mistap never loses the level.
World map or records	Back to the title screen.
Title screen	Android's normal Back (leaves the app).

CHAPTER 4

System Design

SummaRace v3 · version 1.3.6 (versionCode 17) · how the parts of the app fit together and why

4.1 Overview

This document describes the system as a whole: who uses it, its components, how data moves between them, the main runtime sequences, and the design choices behind them. The TRD lists the tools; this document shows how they work together.

In short, **SummaRace is a single-device, offline system.** One Android app holds the game, the content and the saved data. The only data that leaves the phone is a CSV file the teacher exports.

4.2 System context

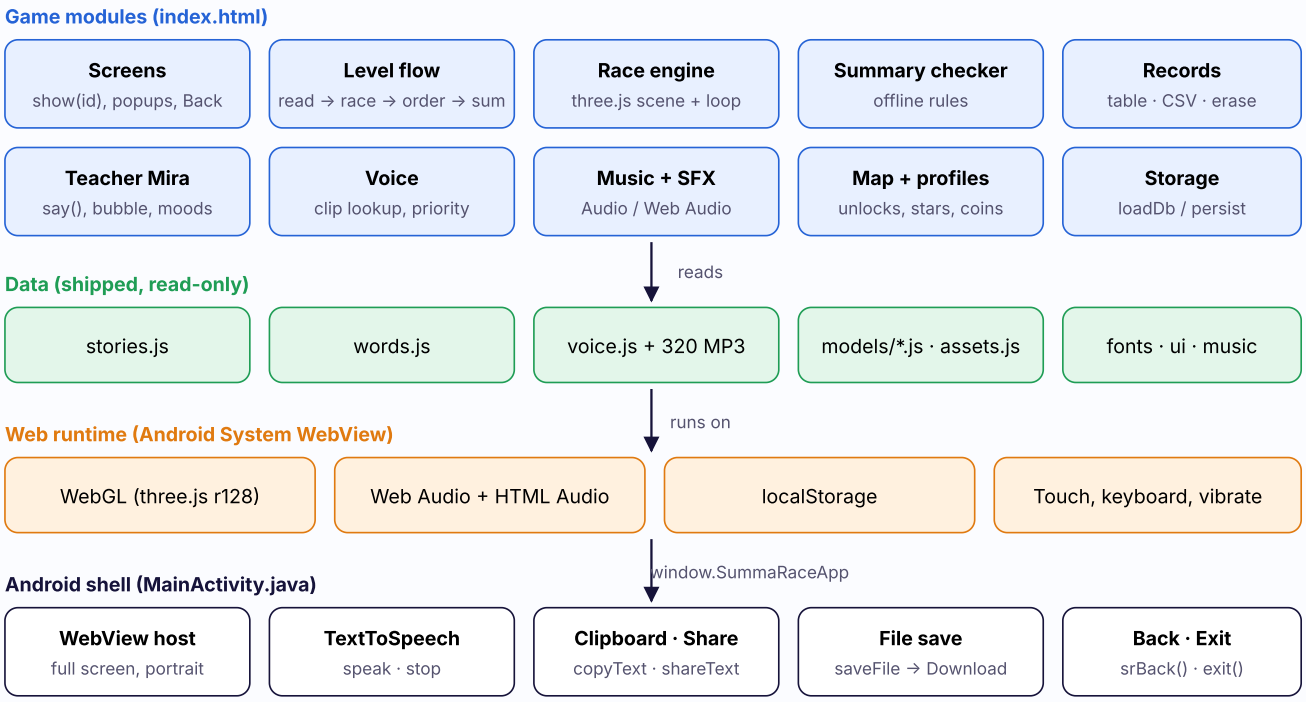
Figure 4.1 System context diagram



Note. The orange arrow is the only data that leaves a phone, and only when the teacher chooses to export it.

4.3 Components

Figure 4.2 Component diagram in layers



Each layer only talks to the one below it. If the Android bridge is missing (desktop browser), the game uses web fallbacks.

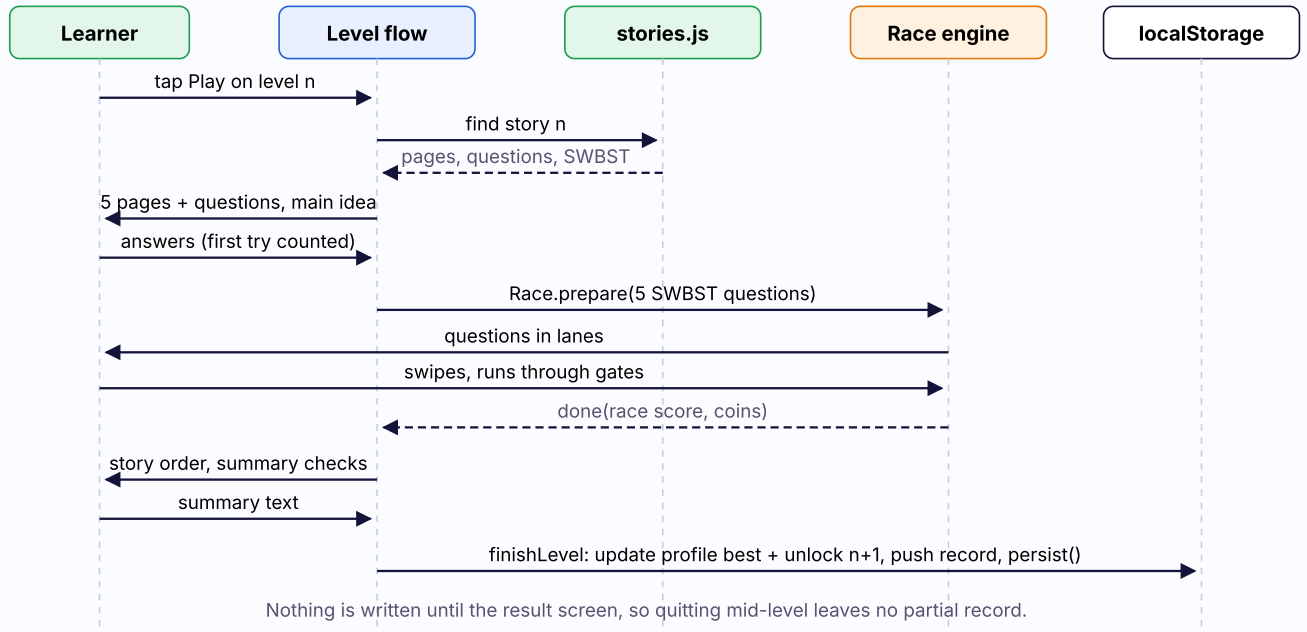
Note. Four layers, top to bottom: game logic, shipped data, the browser engine inside Android, and the thin Java shell.

Table 4.1 Components

Component	Responsibility	Key functions
Screens	Shows one screen at a time, step bar, top bar, popups, Android Back handling.	`show` , `openModal` , `srBack`
Level flow	Runs one level from first page to result and records it.	`startLevel` , `askPage` , `askMain` , `startRace` , `startOrder` , `startSummary` , `finishLevel`
Race engine	3D scene, course, patrol, answer gates, end scenes. Self-contained module with its own state.	`Race.prepare` , `Race.stop` , `Race.warm`
Summary checker	Offline checks and feedback for the 2–3 sentence summary.	`checkSummary` , `scoreSummary` , `meaningChecks`
Teacher Mira + Voice	Shows and speaks lines, picks a recorded clip, never cuts off story reading.	`say` , `line` , `clipFor` , `playClip` , `speakRaw`
Music + SFX	Looping music with fades and ducking; sound effects from decoded clips or synthesised tones.	`Music.play/duck` , `sfx.*`
Map + profiles	World map, level cards, sign-in, unlocks, stars and coins.	`goHome` , `levelCard` , learner form
Records	Teacher table, CSV export, erase with undo.	`goRecords` , `csv` , ERASE / restore
Storage	One JSON object in localStorage, read once, written after every change.	`loadDb` , `persist` , `savePref`

4.4 Data flow through one level

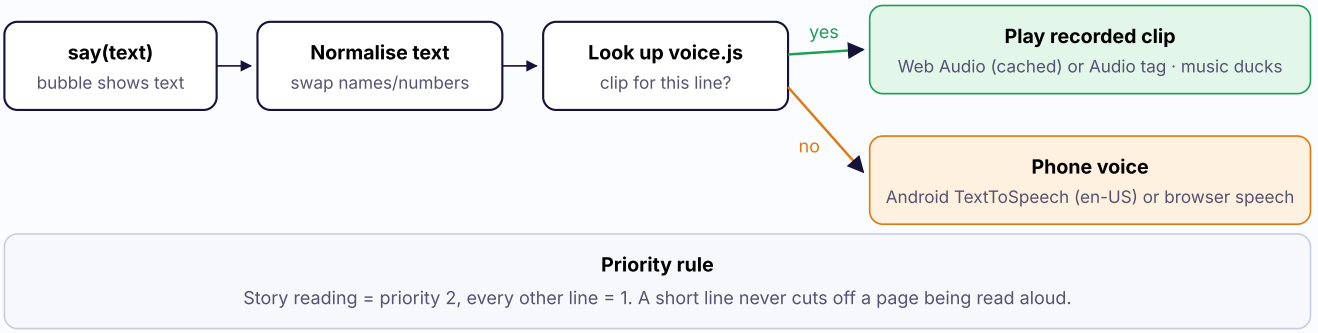
Figure 4.3 Sequence diagram for one level



Note. Solid arrows are calls or actions; dashed arrows are data coming back.

4.5 Teacher Mira's voice pipeline

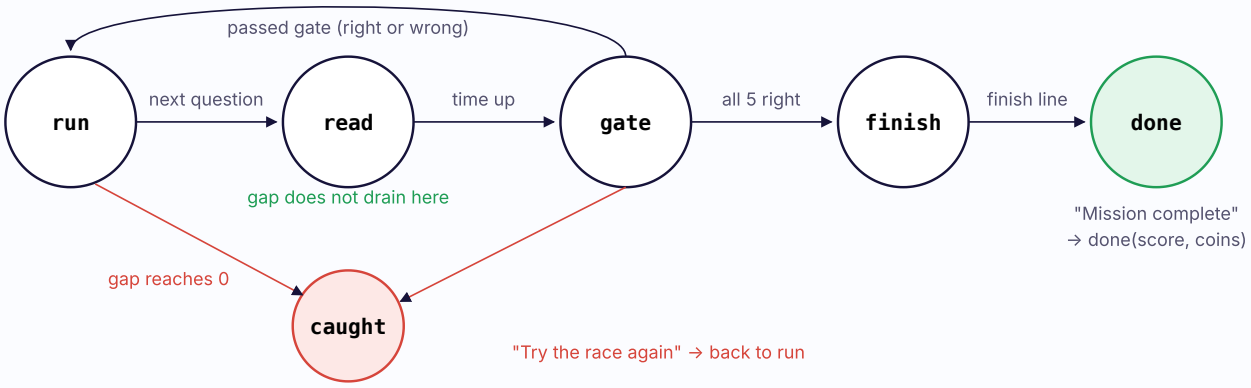
Figure 4.4 Voice pipeline



Note. Voice can be switched off in Settings. Lines without a clip sound robotic because they use the phone's own voice.

4.6 Race engine state machine

Figure 4.5 Race state machine



Note. Each frame: move the world toward the camera at the current speed, update the state, check obstacle and gate collisions, move the patrol by the gap, render.

4.7 Performance design

Table 4.2 Performance design

Problem	Design choice
Slow start-up on cheap phones	Nothing 3D loads at start-up. The 4.5 MB of models preload in the background 1.5 s after the learner's first tap (`Race.warm`), so they are usually ready before the first race.
Low frame rate in the race	The first 2.5 s of a race are timed. Below 22 fps the game switches to low mode: no falling leaves, no sun shadows, less side scenery, pixel ratio 1. Pixel ratio is capped at 1.5 in any case.
Memory growing during a race	Obstacles and effects free their GPU geometry, materials and textures when removed; shared models stay.
Loading files from inside the APK	3D models and sound clips are base64 text inside `.js` files, so they load with a script tag even where file fetches are blocked.
Voice delay	Mira's clips are decoded once and cached in memory.

4.8 Failure handling

Table 4.3 Failure handling

If this fails...	...the game does this
The race cannot start (no WebGL)	Mira says the race could not start and the level continues to Story order with a race score of 50% and 0 coins.
A 3D model does not load	Simple shapes made in code stand in for the runner, patrol or trains.
A voice clip is missing	The phone's voice reads the line instead.
localStorage is blocked or full	Every read and write is wrapped, so the game keeps running; progress is not kept.
Saving the CSV is not allowed (Android 7–9, permission refused)	A copy is kept in the app's own folder and the button tells the teacher to use Share CSV.

If this fails...	...the game does this
Teacher erases by mistake	The last erase is kept and can be brought back, merged with anything recorded since.

4.9 Security and privacy

- No network code, no analytics, no ads, no accounts. The app asks only for vibration and, on Android 7–9, storage (to save the CSV).
- Learner data is a name, a section, scores and summaries, kept only on the phone in the app's private storage.
- Teacher records have no password; ERASE needs a typed word, which guards against accidents, not against learners who deliberately look.
- The repo is public, so the answer key and all content can be read by anyone. The signing key is kept off GitHub.

4.10 Build and release pipeline

Figure 4.6 Build pipeline



Note. There is no CI. Release APKs must be built on the PC that holds the signing key; a build signed with any other key cannot update the installed app.

4.11 Design decisions

Table 4.4 Design decisions

Decision	Why	Trade-off
Web game in a WebView, not a native or Unity app	One code base that runs in a browser for fast testing; small APK (~13 MB); v1/v2 in Unity were heavier.	Depends on the phone's WebView and WebGL.
No backend	Classrooms may have no internet; no accounts or personal data on a server.	Records stay per phone; the teacher must collect CSVs from each device.
Rules-based summary checker	Works offline and gives the same result every time.	Can miss meaning a teacher would catch; the text is saved for the teacher.
Recorded voice instead of live TTS	A natural voice on any phone, no download.	New lines need recording; the build script was lost.
One big `index.html`, no framework or build step	Nothing to install; easy to recover and patch.	A large file to read; modules are separated by comments, not files.
Answers in lanes, no pause (1.3.5+)	Keeps the race moving, per the client's last request.	Slow readers get 4–9 s of reading time before the gate; still worth a phone test with a slow reader.

CHAPTER 5

Design Brief

SummaRace v3 · version 1.3.6 · screenshots taken from `main` at phone size (390 × 844)

5.1 Overview

This document sets the look of SummaRace: colours, fonts, components, and how each screen should look, so new screens match the ones already built.

5.2 Brand identity

Bright, friendly, readable. A cartoon "sticker" world: white and cream cards with thick plum outlines on a bright sky. Big rounded buttons that look pressable. Teacher Mira is always nearby, so the learner never feels alone with a hard question.

For 9–10 year olds

Large text, large tap targets (answers are at least 58 px tall), one task per screen.

Reading first

Story text uses Andika, a font made for early readers. Game chrome never covers the story.

Kind feedback

Wrong answers shake and turn red, but the right answer is never given away. Mira always offers to show the story again.

Game energy

Stars, coins, medals, confetti and a chase scene make finishing a level feel like a win.

5.3 Colour palette

Core

Plum
#3a1460 · outlines, titles

Ink
#2c1a4a · body text

Cream
#fffaf0 · cards

Sky
#2f8cf0 · background

Sun
#ffd23f · stars, highlights

Go

Bad

#22c06c · main buttons, right

#ff4a3d · wrong, close

Story parts (SWBST)

Each story part keeps one colour everywhere: the race, story order and summary feedback.



Somebody
#ff7a1a



Wanted
#ff4f9a



But
#e8352a



So
#1f8fe0



Then
#21b33c

Difficulty



Easy
#1fae4b



Average
#ff9300



Hard
#e8463b

The game has one bright look. It does not switch to a dark theme.

5.4 Typography

Table 5.1 Typography

Role	Font	Used for
Title	Luckiest Guy	Logo, "AMAZING!" / "GREAT JOB!", flash words in the race, score numbers, the story's first letter.
Interface	Fredoka (variable, 300–700)	Buttons, labels, tags, step bar, popups.
Reading	Andika (regular, bold)	Story pages, questions, answers, the summary box.
Back-up	Lilita One, Atkinson Hyperlegible	Listed second in the font stacks; shown only if the main font cannot load.

Headings are white with a thick plum outline on the sky, and plum on cards. Answer text is about 18 px (1.12 rem).

5.5 UI components

Table 5.2 UI components

Component	Look	Use
Main button (`.btn.go`)	Green, rounded, white text, darker bottom edge that presses down 4 px on tap.	The one next step on a screen: Let's go!, Answer the question, Check order.
Second button (`.btn.alt`)	Purple or white with plum outline.	Back, Read the story, Download CSV.
Round icon button (`.cbtn`)	Circle with an icon; red, green, blue, yellow variants.	Home, Settings, Close (red ✕), Listen (yellow), Map / Replay / Next.
Answer option (`.opt`)	White card with a coloured letter badge A/B/C/D. Turns green and pops when right, red and shakes when wrong.	Page questions, main idea.

Component	Look	Use
Race lane chips	Three coloured chips (blue, orange, purple) with A/B/C, one per lane, under the question.	Race questions.
Slot (`.slot`)	Dashed box with a SWBST colour label; solid when filled, green with ✓ when right.	Story order.
Feedback row (`.fbrow`)	Green "✓ Somebody" rows and red "Add: But" rows.	Summary check results.
Popup (`.modal .panel`)	Cream panel with a purple ribbon title and a red close button.	Who is playing?, Settings, Credits, Read it again.
Teacher Mira dock	Mira's picture at the bottom left with a speech bubble; folds to just her face.	Every screen except the race and records.
Step bar (`.steps`)	4 numbered dots: Read & answer, Race, Story order, Summary. Done steps show ✓.	Top of every level screen.

5.6 Screens



Figure 5.1 Title



Figure 5.2 Who is playing?

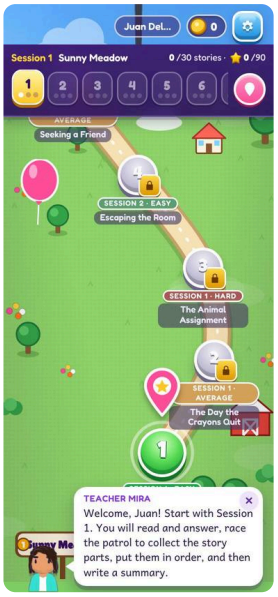


Figure 5.3 World map



Figure 5.4 Read a page

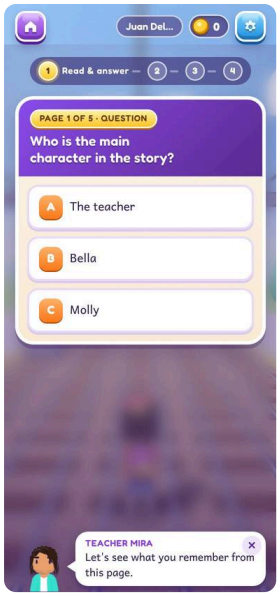


Figure 5.5 Page question

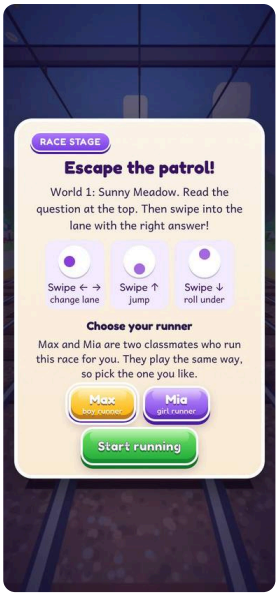


Figure 5.6 Race start card



Figure 5.7 Race: answers in lanes

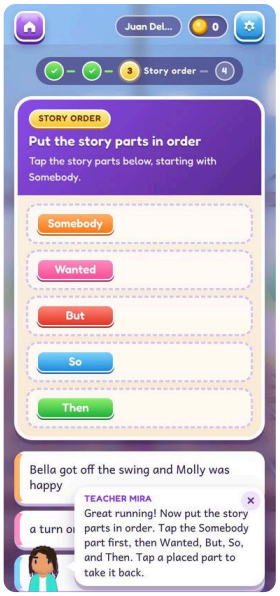


Figure 5.8 Story order



Figure 5.9 Summary feedback

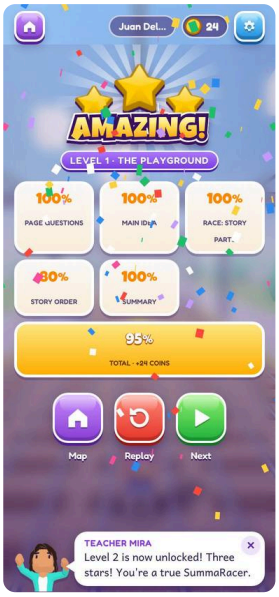


Figure 5.10 Results

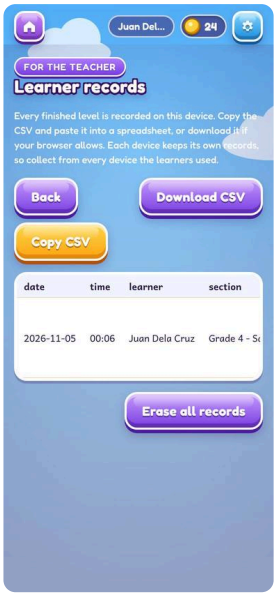


Figure 5.11 Teacher records

Screenshots were taken in headless Chromium, so the 3D race uses software graphics and looks a little softer than on a phone.

5.7 Worlds

Table 5.3 Worlds

World	Levels	Race scenery	Screen backdrop
Sunny Meadow	1–6	Grass, flowers, houses, trees	`ui/world-0.jpg`
River Bend	7–12	River, houses, pines, mist	`ui/world-1.jpg`
Story Town	13–18	City buildings, trees	`ui/world-2.jpg`

World	Levels	Race scenery	Screen backdrop
Palm Beach	19–24	Sand, sea, palms, umbrellas	`ui/world-3.jpg`
Summit Peak	25–30	Snow, mountains, pines	`ui/world-4.jpg`

The race sky cycles through 6 times of day (morning to sunset) by level, so neighbouring levels look different.

5.8 Motion and sound

- Buttons press down on tap. Right answers pop, wrong answers shake. Screens slide in. Confetti on the results screen for 4.5 s.
- If the phone asks for reduced motion, all animations and confetti are turned off.
- Short sound effects for taps, coins, right, wrong, jump, bump and win. Music lowers while Mira talks.
- Voice, sound effects and music each have their own switch in Settings.

5.9 Accessibility

- Every icon button has a text label for screen readers. Mira's bubble is a live region, so her lines are read out.
- Focus outlines are a thick pink ring.
- Answer order is shuffled, so colour or position never tells the answer.
- Every story page can be read aloud with the Listen button.

CHAPTER 6

DATA SCHEMA

SummaRace v3 · version 1.3.6 · this replaces a "backend schema": the game has no backend

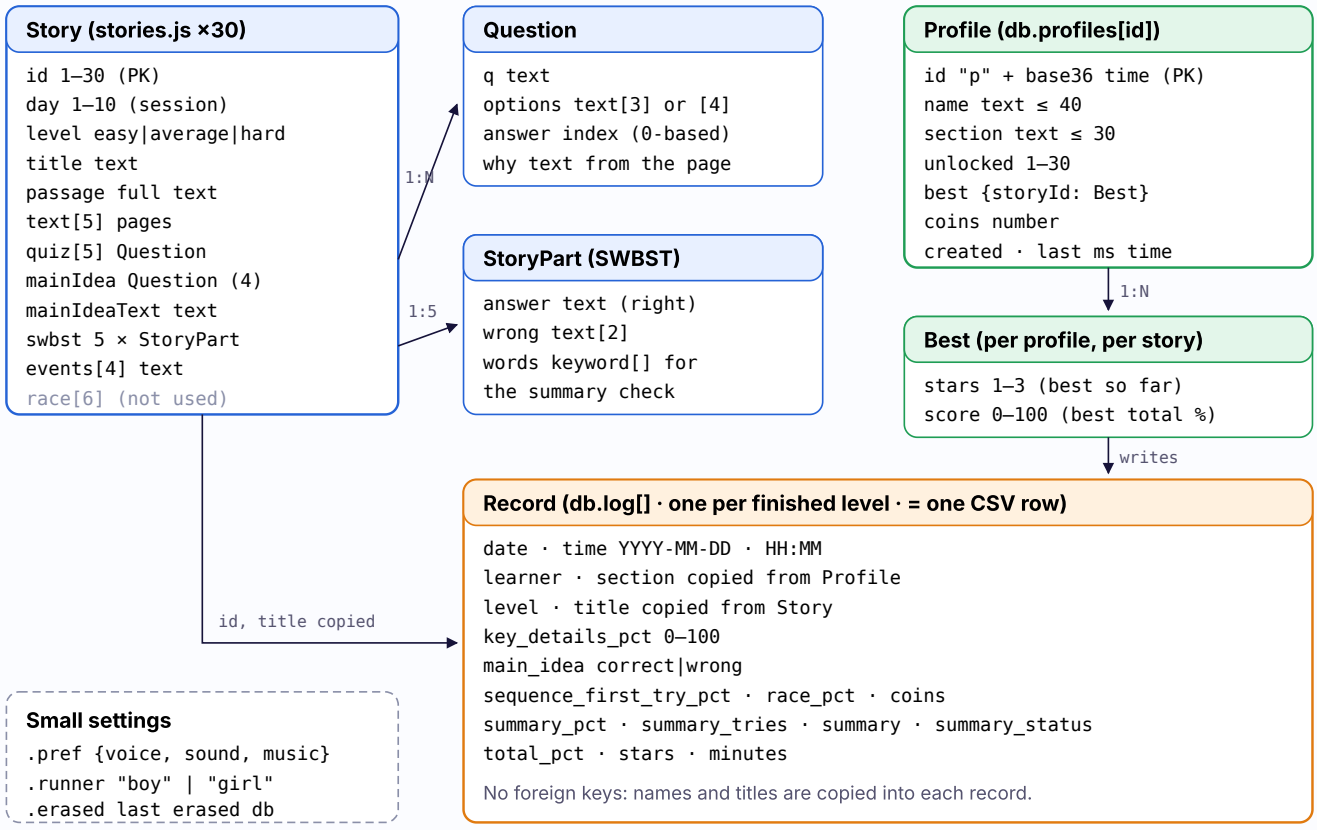
6.1 Overview

This document says where SummaRace's data lives, how it is structured, who can see it, and how the pieces connect.

> **There is no server and no database.** Story content is a JavaScript file shipped in the app. Learner progress and results are saved in the phone's WebView storage (`localStorage`). The only way data leaves the phone is when the teacher copies, downloads or shares the CSV.

6.2 Data model

Figure 6.1 Data model diagram



Note. Blue = shipped content (read-only). Green and orange = learner data saved on the phone. Monospace lines are field name and type.

6.3 Story content: stories.js

`window.STORIES` is an array of 30 objects, sorted by `id` when the game starts. Every story has 5 pages, 5 page questions with 3 options, a main idea question with 4 options, and 5 SWBST parts with 1 right and 2 wrong answers.

Table 6.1 Story content: stories.js

Field	Type	Used for
`id`	number 1–30	Level number and unlock order.
`day`	number 1–10	Session number shown on screens ("Session 1 · Easy").
`level`	`"easy" | "average" | "hard"`	Difficulty tag and colour. Each session has one of each.
`title`	string	Shown everywhere and copied into each record.
`passage`	string	The whole story in one piece. Used by the summary checker (copy check, names, vocabulary).
`text`	string[5]	The 5 pages the learner reads.
`quiz`	Question[5]	One question after each page. `why` is the page text that holds the answer; it is used to highlight the right part when the story is shown again.
`mainIdea`	Question (4 options)	The main idea question after page 5, and the main idea check in the summary.
`mainIdeaText`	string	The main idea in words; allowed vocabulary for the summary.
`swbst`	{somebody, wanted, but, so, then}: StoryPart	Race questions, story order items and summary scoring. `words` are the keywords that prove a summary told that part.
`events`	string[4]	Allowed vocabulary for the summary checker.
`modelSummary`	string	Teacher Mira's example, shown after the 3rd summary try and after a pass.
`race`	Question[6]	**Not used since 1.3.5.** The old trivia race questions. The race now asks the 5 SWBST parts instead.

```
{
  "id": 1, "day": 1, "level": "easy", "title": "The Playground",
  "text": ["Molly and her friend Bella were playing at the playground. …", "…"],
  "quiz": [{ "q": "Who is the main character in the story?",
             "options": ["Bella", "Molly", "The teacher"], "answer": 1, "why": "…"
}, …],
  "swbst": { "somebody": { "answer": "Molly", "wrong": ["Bella", "The teacher"],
                           "words": ["molly", "girl", "kid"] }, … },
  "modelSummary": "Molly wanted her turn on the swing, but Bella would not get off.
…"
}
```

Other shipped data files

Table 6.2 Story fields in stories.js

File	Global	Shape
`words.js`	`SR_WORDS`	One string of 24,887 common English words separated by spaces.
`voice/voice.js`	`SR_VOICE`	Map from normalised line text (lower case, letters and digits only) to a clip path, e.g. `"please type your name first" → "voice/m-001.mp3"`. Story pages use the key `"page <storyId> <page>"`. 170 Mira lines + 150 pages = 320 clips.

File	Global	Shape
`assets.js`	`SR_ASSETS`	`{textures:{platformer}, models:{coin, cloud}, sounds:{bad, bump, click, coin, good, hit, jump, land, swish, tap, win}}` as data URLs / base64.
`models/*.js`	`SR_MODELS`	`SR_MODELS[name] = "<base64 GLB>"` for boy, girl, officer, city, suburb, trains, nature.

6.4 Saved data: localStorage

Table 6.3 Other shipped data files

Key	Value	Written when
`summarace.v1`	`{profiles: {id: Profile}, current: id | null, log: Record[]}`	A learner signs in, finishes a level, or the teacher erases or restores.
`summarace.v1.pref`	`{voice: bool, sound: bool, music: bool}` (all true by default)	A switch in Settings changes.
`summarace.v1.runner`	`"boy"` (Max) or `"girl"` (Mia)	The runner is picked on the race start card.
`summarace.v1.erased`	A full copy of `summarace.v1` from just before the last erase	The teacher types ERASE. Removed again by "Bring back erased records", which merges it with anything recorded since.

An unfinished level is never saved: quitting mid-level leaves no record. If storage is blocked or full, the game keeps working for the session but nothing is kept.

6.5 CSV export format

File name `summarace-records-YYYY-MM-DD.csv`, UTF-8 with a byte-order mark (so Excel opens it correctly), comma separated, one header row, then one row per finished level, oldest first. Values with a comma, quote or line break are wrapped in double quotes. The column order is fixed:

Table 6.4 Keys saved in localStorage

#	Column	Type	Meaning
1	`date`	YYYY-MM-DD	Date the level was finished (UTC date).
2	`time`	HH:MM	Local time on the phone.
3	`learner`	text	Name as typed.
4	`section`	text	Section or group as typed (may be empty).
5	`level`	1–30	Story id.
6	`title`	text	Story title.
7	`key_details_pct`	0–100	Page questions right on the first try.
8	`main_idea`	correct | wrong	Main idea right on the first try.
9	`sequence_first_try_pct`	0–100	Story parts in the right slot on the first "Check order".
10	`race_pct`	0–100	Story parts answered right on the first try in the race.
11	`coins`	number	Coins picked up in the race.
12	`summary_pct`	0–100	Story parts found in the passing summary, out of 5, minus half a part (10 points) for each wrong detail. 0 if handed to the teacher.

#	Column	Type	Meaning
13	`summary_tries`	number	Scored summary checks (early checks like "too short" do not count).
14	`total_pct`	0–100	Average of read & answer, race, order and summary.
15	`stars`	1–3	Stars for this attempt (≥85 → 3, ≥65 → 2, else 1).
16	`minutes`	number, 1 decimal	Time from opening the level to the result.
17	`summary`	text	The learner's summary, word for word.
18	`summary_status`	passed | passed after example | teacher check	How the summary step ended. Added in 1.3.2, kept as the last column so old sheets still line up.

```
date,time,learner,section,level,title,key_details_pct,main_idea,sequence_first_try_p
ct,race_pct,coins,summary_pct,summary_tries,total_pct,stars,minutes,summary,summary_
status
2026-11-05,08:06,Juan Dela Cruz,Grade 4 - Sampaguita,1,The Playground,100,correct,8
0,100,24,100,1,95,3,6.4,"Molly wanted a turn on the swing, but Bella would not get o
ff. …",passed
```

> Column `story_elements_first_try_pct` existed before 1.3.3 and was a copy of `race_pct`. It was removed, so sheets from older builds have one more column.

6.6 Access control

Table 6.5 CSV columns

Who	Story content	Own progress	Other learners	Records / CSV
Learner	Read	Read, add by playing	Sees names in "Or continue as"	Can open Teacher records (no password)
Teacher	Read	—	Sees all on the phone	Copy, download, share, erase (type ERASE), restore
Anyone with the APK	Read (the files are inside the app; the repo is public)	Nothing: data stays on each phone		

There is no password on Teacher records. This was a choice for a classroom tool; the ERASE word and the undo protect against accidents, not against a learner who deliberately looks.

6.7 Data storage

Where

WebView localStorage inside the app's private folder. Clearing the app's data or uninstalling deletes it.

Export

Android 10+: Download/SummaRace via MediaStore. Android 7–9: Download/SummaRace with storage permission. Or Share / Copy.

Backups

None automatic. The teacher should export the CSV after each session.

Personal data

Only the name and section typed in, and the summaries. Nothing is sent anywhere.

CHAPTER 7

Implementation Plan

SummaRace v3 · written after the build, as a record of what was built and in what order

7.1 Overview

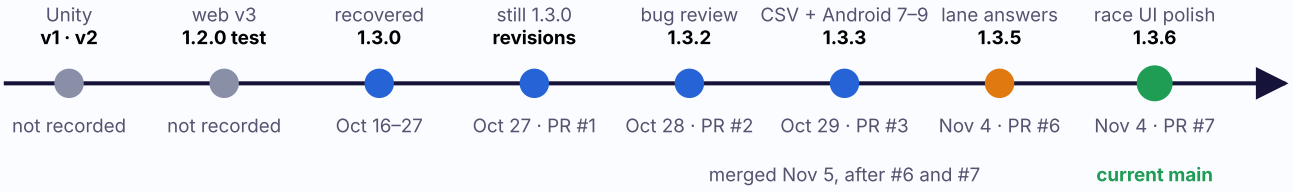
This document breaks SummaRace v3 into the phases it was actually built in, with what each phase delivered. Dates from 2026-10-27 on come from the git history. Earlier dates were not recorded and are marked "not recorded".

7.2 Goals

Deliver an offline Android reading game for the research team's 30 stories, with records the teacher can export, built and tested in small steps so each client revision could be checked on a phone before it was merged.

7.3 Timeline

Figure 7.1 Timeline of releases



Note. The client revisions kept versionName 1.3.0 on main (a 1.3.1 test APK of that period is in the project files). 1.3.4 was a local test build only (PR #3 plus PR #5) and was never merged.

7.4 Implementation phases

Table 7.1 Implementation phases

Phase	What got built	Key deliverables	When
0. Earlier versions	SummaRace v1 and SummaRace2 in Unity.	Separate repos (`orbitsdev/SummaRace` , `orbitsdev/SummaRace2`). v3 is a rewrite, not a port.	Not recorded
1. Web game v3	The game as an HTML/JS app with three.js: title, profiles, world map, read and quiz, story order, SWBST, summary, results, records, 3D race, Teacher Mira's recorded voice, 30 stories.	SummaRace 1.2.0 test APK.	Not recorded

Phase	What got built	Key deliverables	When
2. Recovery	The original project folder was lost. The web game was taken back out of the 1.2.0 APK and a new Android wrapper was written.	Repo created with the game in `app/src/main/assets/www`, `MainActivity.java` with the bridge, new signing key, 1.3.0. Lost for good: `build-voice.sh`, `CREDITS.md`, the old key.	2026-10-16 to 2026-10-27
3. Client final revisions	The client's revision list.	Race questions at a "Story stop" (later replaced), runner picker explained, one story-order instruction with a 30 s limit after a wrong try, summary in 2–3 sentences, offline content checks for each story part, wrong details, unknown names, "not", order and swapped characters.	2026-10-27 · PR #1
4. Bug review fixes	A 19-point review of the game, top items fixed.	Shuffled answers, Back button safety, Mira bubble folding, race GPU clean-up, device voice fallback, ERASE + undo, copy-check fix, "my teacher will check" + `summary_status` column, softer race penalty, order timer by Check, viewport and font fixes. 1.3.2 (versionCode 13).	2026-10-28 · PR #2
5. Review fixes 2	Two more review items.	Dropped the duplicate CSV column, CSV saves to Download/SummaRace on Android 7–9. 1.3.3 (versionCode 14).	2026-10-29 · PR #3 (merged 2026-11-05)
6. Lane answers	The client's last request: the race no longer stops for a question.	Question at the top, answers A/B/C in the 3 lanes, run through the right lane's gate. Reading time before each gate. 1.3.5 (versionCode 16).	2026-11-04 · PR #6
7. Race UI polish	Make the race readable on a phone.	Slim question panel, solved count in the top bar, Mira hidden during questions, centred start card with a swipe tutorial, on-screen arrow buttons removed. 1.3.6 (versionCode 17).	2026-11-04 · PR #7
8. Documentation	These documents and the full asset list.	`docs/`: PRD, TRD, App Flow, Design Brief, Data Schema, this plan, Assets & Credits.	2026-11-05

7.5 How each phase was run

Figure 7.2 Work loop



Note. PR #6 and PR #7 were merged before a full phone test, at the owner's request. A 1.3.6 test APK was built for that test.

7.6 Milestones

Table 7.2 Milestones

Milestone	Description	Status
Source recovered	Game runs from the repo in a browser and builds as an APK.	Done 2026-10-27
Client revisions applied	All items from the client's final revision list.	Done 2026-10-27
Review fixes	Top items of the 19-point review.	Done 2026-10-28/29

Milestone	Description	Status
Race answers in lanes	Client's last request.	Done 2026-11-04
Documentation	This set of documents.	In review
Release build 1.3.6	Signed with the project key on the owner's PC, tested on a phone, handed to the client.	Owner to confirm

7.7 What is left

- Decide on PR #4 (race music speeds up) and close PR #5 (no longer needed).
- Ask the client about the two stories titled "In Grandfather's Day" (levels 15 and 18).
- Record voice clips for the newer Mira lines, or keep reusing recorded lines, since the clip build script was lost.
- Phone checks still worth doing: a phone with a notch, a 2 GB phone, the on-screen keyboard over the summary box, and a slow reader in the race.
- Next release must use versionCode 18 or higher.

CHAPTER 8

System Requirements and Setup

SummaRace v3 · release 1.3.7 (versionCode 18) · what a laptop, PC or phone needs to run the game, and how to set it up

8.1 Overview

SummaRace is made for Android phones, but the game itself is a web page, so it also runs on a laptop or desktop PC in a normal web browser. This document lists what each kind of device needs and gives the steps to get the game running.

Play on a laptop or PC

Open the game in Chrome or Edge. No install and no internet needed. Good for demos, panel defences and teachers checking the stories. See sections 2–4.

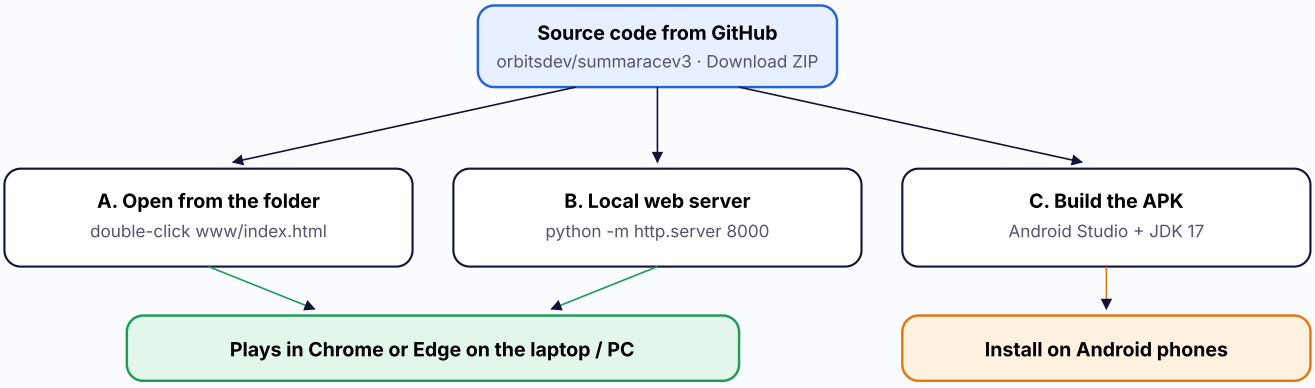
Play on an Android phone

Install the APK. This is how learners use it in class. See section 5.

Build a new version

For a developer changing the game: Android Studio, JDK 17 and the signing key. See section 6.

Figure 8.1 Three ways to run SummaRace



Note. Options A and B need no install other than a browser (and Python for B). Option C is only for making a new APK.

8.2 Laptop or PC requirements

Table 8.1 Laptop or PC requirements

Item	Minimum	Recommended
Operating system	Windows 10, macOS 11, or a recent Linux desktop	Windows 10 or 11 (64-bit)

Item	Minimum	Recommended
Web browser	Chrome or Edge 80, Firefox 74, Safari 13.1 (the code uses modern JavaScript such as `?.`)	The latest Google Chrome or Microsoft Edge
3D graphics	WebGL turned on in the browser (any graphics chip from the last 10 years)	Hardware acceleration on in the browser settings
Processor and memory	Dual-core CPU, 4 GB RAM	Quad-core CPU, 8 GB RAM
Disk space	About 20 MB for the game folder (the `www` folder is 18 MB)	About 100 MB if the whole repository ZIP is kept
Screen	1280 × 720	1920 × 1080. The game also fits a narrow window or the browser's phone view.
Sound	Speakers or headphones for Teacher Mira's voice and the music	—
Input	Mouse or touchpad and keyboard	A touch screen also works (swipe like on a phone)
Internet	Only to download the files once. The game itself never goes online.	—
Python (option B only)	Python 3.8 or newer, for the local web server	The latest Python 3 from python.org

> Tested on 11 November 2026: the full game (title, reading, questions, 3D race, story order, summary, results and records) runs in Chromium both straight from the folder (option A) and from a local server (option B).

8.3 Install guide: play on a Windows laptop or PC

Each step says **what** to do, **where** to do it, the **command** to type (if any) and what you **should see** when it worked. Option A needs only a browser. Option B also needs Python.

1. **Check the browser**
   Where: Google Chrome or Microsoft Edge (Edge is already on every Windows 10/11 PC).
   Type in the address bar: `chrome://settings/help` (Chrome) or `edge://settings/help` (Edge), then press Enter.
   You should see: "Chrome is up to date" or "Microsoft Edge is up to date" and a version number of 80 or higher.
   If Chrome is missing: download it from google.com/chrome.

2. **Turn on graphics acceleration (needed for the 3D race)**
   Type in the address bar: `chrome://settings/system` or `edge://settings/system`
   Do: switch on **Use graphics acceleration when available** and click **Relaunch** if it asks.

3. **Download the game**
   Where: in the browser, open github.com/orbitsdev/summaracev3/archive/refs/heads/main.zip (or open github.com/orbitsdev/summaracev3 and click the green **Code** button → **Download ZIP**).
   You should see: `summaracev3-main.zip` in your **Downloads** folder.

4. **Unzip it**
   Where: File Explorer → **Downloads**. Right-click `summaracev3-main.zip` → **Extract All...** → choose a folder such as `C:\Users\<you>\Documents` → **Extract**.
   You should see: a folder `Documents\summaracev3-main`. The game itself is in `summaracev3-main\app\src\main\assets\www` (it contains `index.html`, `stories.js` and the `models`, `voice` and `fonts` folders).

Option A: open the game straight from the folder (no install)

1. **Open the game**
   Where: File Explorer → `summaracev3-main\app\src\main\assets\www`
   Do: right-click `index.html` → **Open with** → **Google Chrome** or **Microsoft Edge**.
   You should see: the SummaRace title screen with Teacher Mira and a green **PLAY** button. The address bar starts with `file:///C:/`.

2. **Go full screen** with **F11** (press F11 again to leave), then click **PLAY**.

Option B: run it from a local web server (for developers)

1. **Install Python 3**

   Download: python.org/downloads/windows → latest **Python 3** Windows installer (64-bit).

   Do: run the installer and **tick "Add python.exe to PATH"** on the first screen, then **Install Now**.

2. **Check that Python works**

   Where: press the **Windows key**, type `cmd`, press Enter (Command Prompt opens).

   Type: `python --version`

   You should see: `Python 3.x.x`. If it says "not recognized", see section 7.

3. **Open a Command Prompt inside the game folder**

   Where: File Explorer → `summaracev3-main\app\src\main\assets\www`. Click the **address bar** at the top, type `cmd` and press Enter.

   You should see: a Command Prompt whose line ends with `...\app\src\main\assets\www>`

4. **Start the server**

   Type: `python -m http.server 8000`

   You should see: a line like `Serving HTTP on :: port 8000 ...`. Leave this window open while playing.

5. **Open the game**

   Where: Chrome or Edge address bar.

   Type: `http://localhost:8000`

   You should see: the title screen. To stop the server, click the Command Prompt window and press **Ctrl + C**, or close it.

Phone-sized view (optional)

In Chrome or Edge press **F12**, click the phone-and-tablet icon at the top left of the panel (or press **Ctrl + Shift + M**), pick a phone such as *iPhone 12 Pro* (390 × 844), then press **F5** to reload.

8.4 Playing on a laptop or PC

Table 8.2 Playing on a laptop or PC

Action	Keyboard	Mouse / touch
Change lane (pick an answer in the race)	← → or A / D	Click the left or right side of the track, or swipe
Jump	↑, W or Space	Swipe up
Roll	↓ or S	Swipe down
Close a popup	Esc	The close button

Differences from the phone app

- **Saved progress and records stay in that browser** on that computer. Another browser, a private window, or "clear browsing data" starts empty. Option A and option B also keep separate records.
- **Records:** use **Download CSV** (saved to the browser's Downloads folder) or **Copy CSV**. Share may not be offered on a PC.
- **Voice:** Mira's recorded clips play as normal. The few lines without a clip use the computer's built-in voice.
- **Exit game** is hidden; close the browser tab instead.
- The browser may block sound until the first click. This is normal; sound starts after **PLAY**.

8.5 Android phone requirements

Table 8.3 Android phone requirements

Item	Requirement
Android version	Android 7.0 (API 24) or newer; built for Android 15 (API 35)
Memory	About 2 GB RAM or more
Storage	About 50 MB free for the app and its saved records
Screen	Portrait phone screen; phones with a notch are supported
Internet	Not needed
Permissions	Vibration; on Android 7–9 only, storage (to save the CSV to Download/SummaRace)

Before installing 1.3.7: export records, then uninstall the old app

> **Important.** Version 1.3.7 (versionCode 18) is signed with a **new key**. It cannot update any earlier SummaRace on a phone: the 1.2.0 test build and every 1.3.x build, **including 1.3.6**, must be **uninstalled first**. Uninstalling **deletes all learner progress and records** on that phone, and the game cannot import them back. Do these steps on every phone first:

1. **Export the records**
   Where: on the phone, open SummaRace → **PLAY** → **Teacher records** (or Settings → Teacher records).
   Do: tap **Download CSV**. Also tap **Share CSV** and send it to yourself (email or Drive) as a second copy.
   You should see: `summarace-records-YYYY-MM-DD.csv` in the phone's **Download/SummaRace** folder. Copy it to a PC before going on.
2. **Uninstall the old app**
   Where: press and hold the SummaRace icon → **App info** (or Settings → Apps → SummaRace) → **Uninstall** → **OK**.
   Note: learners start again from level 1 in 1.3.7. Their earlier results stay only in the exported CSV.

Install the APK

1. **Copy the APK to the phone:** connect the phone with a USB cable, choose **File transfer** on the phone, then in File Explorer drag `app-release.apk` into the phone's **Download** folder. (Google Drive or a messaging app also works.)
2. **Open it on the phone: Files** app → **Downloads** → tap `app-release.apk`. When Android asks, allow **Install unknown apps** for the Files app, go back and tap **Install**.
3. **You should see:** "App installed" and a SummaRace icon on the home screen. Open it and play the first page to check it works.

8.6 Install guide: build a new APK (developers)

Table 8.4 Install guide: build a new APK (developers)

Tool	Version	Where to get it
Android Studio	Ladybug (2024.2) or newer	developer.android.com/studio. It includes the JDK and installs the Android SDK.
JDK	17 (tested)	Comes inside Android Studio. For the command line, install it yourself (Way 2, step 1).
Android Gradle Plugin	8.7.3	Downloaded automatically on the first sync (set in `build.gradle`).
Gradle	8.9 or newer (tested: 8.14)	The repository has no Gradle wrapper. Install Gradle yourself (Way 2, step 4) or let Android Studio set it up.

Tool	Version	Where to get it
Android SDK Platform	Android 15 (API 35)	Android Studio → SDK Manager (step 3).
Git (optional)	Any	git-scm.com/download/win, or use GitHub Desktop, or the ZIP from section 3.
PC	8 GB RAM, about 10 GB free disk	—
Signing key	The SummaRace release key (`.jks`) and its passwords	The new key made for 1.3.7 (11 November 2026), kept only on the owner's PC, never in the repository. Only needed to build a release APK, not to read these docs or play on a PC. To make a new one, see the end of this section.

There are two ways to build: **Way 1, Android Studio** (easier, but these steps have **not yet been tested** end to end on this project) and **Way 2, the command line** (tested: it built 1.3.7).

Way 1: build in Android Studio (not yet tested)

1. **Get the source code**
   Either: download and unzip the ZIP as in section 3, steps 3–4.
   Or with Git: open Command Prompt in `Documents\GitHub` and type `git clone https://github.com/orbitsdev/summaracev3.git`
   You should see: a folder that contains `settings.gradle` , `build.gradle` and the `app` folder. This is the **project folder**.

2. **Install Android Studio**
   Download: developer.android.com/studio → run the installer with the default options.
   First start: the Setup Wizard opens. Choose **Standard**, accept the licences and click **Finish**. It downloads the Android SDK (this takes a while).

3. **Install Android 15 (API 35)**
   Where: Android Studio → **Tools → SDK Manager** (on the Welcome screen: **More Actions → SDK Manager**) → tab **SDK Platforms**.
   Do: tick **Android 15.0 ("VanillaIceCream") API 35** → **Apply** → **OK**.

4. **Open the project**
   Where: Android Studio → **File → Open** (Welcome screen: **Open**) → select the **project folder** from step 1 → **OK**. Choose **Trust Project** if asked.
   Wait: for the Gradle sync to finish (progress bar at the bottom right). Check **View → Tool Windows → Build**.
   You should see: `BUILD SUCCESSFUL` or "Gradle sync finished", and **app** in the Project panel on the left.
   If it says Gradle or the wrapper is missing: open **File → Settings → Build, Execution, Deployment → Build Tools → Gradle**, check **Gradle JDK** is 17 or newer, and accept Android Studio's offer to set up Gradle (8.9 or newer). Then **File → Sync Project with Gradle Files**.

5. **Raise the version number**
   Where: Project panel → `app` → `build.gradle` (Module :app).
   Change: raise `versionCode` by 1 (after 1.3.7 / 18, the next is `versionCode 19`) and set `versionName` to the new version, e.g. `'1.3.8'` . Then click **Sync Now** at the top of the file.

6. **Test build (optional, not for learners)**
   Where: **Build → Build App Bundle(s) / APK(s) → Build APK(s)**.
   You should see: a pop-up "APK(s) generated successfully" with a **locate** link. The file is `app\build\outputs\apk\debug\app-debug.apk` . A debug APK cannot update the release app on the phones.

7. **Release build, signed with the SummaRace key**
   Where: **Build → Generate Signed App Bundle or APK…** → choose **APK** → **Next**.
   Key store path: **Choose existing…** → the `.jks` file in `Documents\SummaRace-signing` → enter the key store password, key alias and key password → **Next**.
   Build variant: **release** → **Create** (Finish).
   You should see: "APK(s) generated successfully" → click **locate**. The file is `app\release\app-release.apk` in the project folder.

8. **Install it on a phone** as in section 5, then play one full level before handing it out.

Never commit the keystore, `*.jks`, `*.keystore` or `local.properties`. The repository is public. A build signed with any other key cannot update the app already on the phones.

Way 2: build from the command line (tested)

Tested on 11 November 2026: version 1.3.7 (versionCode 18) was built this way on Windows with **JDK 17**, **Android SDK Platform 35** and **Gradle 8.14**, with no Gradle wrapper.

1. **Install JDK 17**
   Download: adoptium.net → Temurin **17**, Windows x64, **.msi**. In the installer, turn on **Set JAVA_HOME** and **Add to PATH**.
   Check in a new Command Prompt: `java -version`
   You should see: `openjdk version "17.…"`

2. **Get the Android SDK (Platform 35 and Build-Tools)**
   Easiest: install Android Studio once (steps 2–3 of Way 1); it puts the SDK in `C:\Users\<you>\AppData\Local\Android\Sdk`. In SDK Manager, tab **SDK Tools**, also tick **Android SDK Build-Tools** (35.x).
   Without Android Studio: download the **Command line tools only** ZIP from developer.android.com/studio, then run `sdkmanager "platforms;android-35" "build-tools;35.0.0"`
   You should see: folders `Sdk\platforms\android-35` and `Sdk\build-tools\35.0.0` (your Build-Tools number may differ; use the folder you have in the commands below).

3. **Tell the build where the SDK is**
   Where: Windows key → type `environment variables` → **Edit the system environment variables** → **Environment Variables…** → under *User variables* click **New**.
   Add: name `ANDROID_HOME`, value `C:\Users\<you>\AppData\Local\Android\Sdk` → **OK**. Open a **new** Command Prompt afterwards.
   Or: create `local.properties` in the project folder with the line `sdk.dir=C\:\\Users\\<you>\\AppData\\Local\\Android\\Sdk`. This file is in `.gitignore`; never commit it.

4. **Install Gradle 8.14** (the repository has no Gradle wrapper)
   Download: gradle.org/releases → **v8.14** → **binary-only**. Unzip to `C:\Gradle`, so that `C:\Gradle\gradle-8.14\bin` exists.
   Add to PATH: in the same Environment Variables window, select **Path** → **Edit** → **New** → `C:\Gradle\gradle-8.14\bin` → **OK**.
   Check in a new Command Prompt: `gradle --version`
   You should see: `Gradle 8.14` and a line with JVM 17.

5. **Raise the version number**
   In `app\build.gradle` (any text editor), raise `versionCode` and `versionName` as in Way 1, step 5.

6. **Build the release APK (unsigned)**
   Where: Command Prompt in the **project folder** (File Explorer → project folder → address bar → `cmd` → Enter).
   Type: `gradle assembleRelease`
   You should see: `BUILD SUCCESSFUL` and the file `app\build\outputs\apk\release\app-release-unsigned.apk`. The first build downloads the Android Gradle Plugin, so it needs internet and takes a few minutes.

7. **Align the APK (zipalign)**
   Type (same window, one line):

```
"%ANDROID_HOME%\build-tools\35.0.0\zipalign.exe" -p -f 4 app\build\outputs\apk\release\app-release-unsigned.apk app-release-aligned.apk
```

   You should see: no error, and `app-release-aligned.apk` in the project folder.

8. **Sign it (apksigner)**

   Type (one line; change the key path, alias and version to yours):

   ```
   "%ANDROID_HOME%\build-tools\35.0.0\apksigner.bat" sign --ks "C:\Users\<you>\Documents\SummaRace-signing\summarace-release.jks" --ks-key-alias summarace --out SummaRace-1.3.7.apk app-release-aligned.apk
   ```

   Then: type the key store password when asked (nothing shows while typing; that is normal).

   You should see: `SummaRace-1.3.7.apk` in the project folder. This is the file to install on phones.

9. **Check the signature**

   Type: `"%ANDROID_HOME%\build-tools\35.0.0\apksigner.bat" verify --print-certs SummaRace-1.3.7.apk`

   You should see: `Signer #1 certificate DN: …` with the names you gave the key, and no error.

> Always zipalign **before** apksigner. Signing first and aligning afterwards breaks the signature.

Create a new signing key (only when the current key is not available)

> **Read this first.** Android only lets an app update itself when the new APK is signed with the **same key**. An APK signed with a new key **cannot update** SummaRace already on a phone: the old app must be **uninstalled first**, and uninstalling **deletes every learner's progress and records on that phone**. Before switching keys, the teacher must **Download CSV** from Teacher records on every phone.

1. **Make a folder for the key, outside the project folder**

   Where: File Explorer → `Documents` → right-click → **New** → **Folder** → name it `SummaRace-signing`.

   Why: anything inside the project folder can end up on GitHub by mistake.

2. **Easiest way: create the key in Android Studio**

   Where: **Build** → **Generate Signed App Bundle or APK…** → **APK** → **Next** → under Key store path click **Create new…**

   Fill in: Key store path `C:\Users\<you>\Documents\SummaRace-signing\summarace-release.jks`, a password (twice), Alias `summarace`, a key password, Validity `25` years, and at least one name field (for example Organization) → **OK**.

   Then: continue with Way 1, step 7 (release → Create), or use the new `.jks` in Way 2, step 8.

3. **Or create it with a command (keytool)**

   Where: Command Prompt opened in `Documents\SummaRace-signing` (address bar → `cmd` → Enter).

   Type (one line; keytool comes with Android Studio):

   ```
   "C:\Program Files\Android\Android Studio\jbr\bin\keytool.exe" -genkeypair -v -keystore summarace-release.jks -alias summarace -keyalg RSA -keysize 2048 -validity 10000
   ```

   Answer: the password questions and the name questions, then type `yes` to confirm.

   You should see: `[Storing summarace-release.jks]`, and the file `summarace-release.jks` in the folder.

4. **Write down and back up**

   Keep: the `.jks` file, the key store password, the alias (`summarace`) and the key password.

   Back up: copy the `.jks` to a USB drive and to private cloud storage. If it is lost again, every phone will need another uninstall.

5. **Never commit it**

   Do not put `*.jks`, `*.keystore`, passwords or `local.properties` in the project folder or on GitHub; the repository is public. The project's `build.gradle` has no signing settings, so nothing in the code needs to change: the key is picked in the step 7 wizard each time.

8.7 Troubleshooting

Table 8.5 Troubleshooting

Problem	Fix
Blank or white page	Use Chrome or Edge, and check the whole `www` folder was copied (not only `index.html`). If it still fails, use option B.
"The race could not start"	WebGL is off. In Chrome or Edge, turn on **Settings → System → Use graphics acceleration**, then restart the browser.
The race is slow or jerky	Close other tabs and plug in the laptop. The game switches to a lighter mode by itself below 22 frames per second.
No sound	Click once on the page, check the volume, and check the Voice, Sound and Music switches in Settings.
Progress is gone	The browser's data was cleared, or a different browser, private window or option (A or B) is in use.
`python` is not recognised	Install Python 3 from python.org and tick **Add python.exe to PATH**, or try `py -m http.server 8000`.
Port 8000 is busy	Use another number, for example `python -m http.server 8080` and open http://localhost:8080.

CHAPTER 9

Handover and open items

State of the project on 10 November 2026

9.1 What is handed over

Table 9.1 Items handed over

Item	Where
Source code (game and Android app)	github.com/orbitsdev/summaracev3, branch `main`, version 1.3.6 (versionCode 17)
Documentation	This package (HTML and PDF), and the same chapters as separate pages in `docs/`
Answer key	Appendix B, built from the current `stories.js`
How to run it on a laptop, PC or phone	Chapter 8: requirements, setup steps, controls and troubleshooting
Release signing key	A new key was made on 11 November 2026 for release 1.3.7 (versionCode 18). It is kept only on the owner's PC, never in the repository. Every future update must be signed with this same key, or phones cannot update the installed app. Phones with any earlier build must export records and uninstall first (Chapter 8, section 8.5).
Next version number	versionCode 19 or higher (1.3.7 used 18)

9.2 Open items

Table 9.2 Open items at handover

#	Item	Why it matters	Owner	Status
1	Play through the signed 1.3.7 release on a real phone	The lane-answer race (PR #6) and the race UI changes (PR #7) were merged before a full phone test.	Developer	To do
2	Confirm the source of the music and all UI images	Needed for honest credits (Appendix A, section A.2).	Owner	To confirm
3	Credit wordfreq (CC BY-SA 4.0) on the credits screen	Its license requires a credit line.	Developer	To do
4	Fix the credits screen link to the lost CREDITS.md	Point it to this package or ship a new credits file.	Developer	To do
5	Ship the font license texts (OFL, Apache 2.0)	Both licenses ask for the text to travel with the fonts.	Developer	To do
6	Confirm story permissions and author credits	Several stories are published books.	Client	To confirm
7	Two stories titled "In Grandfather's Day" (levels 15 and 18)	Confusing on the map and in the CSV.	Client	To confirm
8	PR #4 (race music speeds up)	Optional feature.	Owner	Owner to decide
9	PR #5 (Story Stop voice lines)	No longer needed since the race uses lane answers.	Owner	Close
10	10+ Mira lines have no voice clip	They use the phone's built-in voice. The clip build script was lost.	—	Known limit

11	Same name and section on one phone share a profile	Two learners with the same name in one section would mix their progress.	—	Known limit

REFERENCE

GLOSSARY

APK	Android Package: the installable file of an Android app.
versionCode / versionName	Android version numbers. versionCode (17) must rise with every update; versionName (1.3.6) is the one people see.
WebView	The browser engine an Android app uses to show web pages. SummaRace runs inside one.
Bridge (SummaRaceApp)	The small set of Java functions the game can call: speak, stop, copy, share, save file and exit.
three.js	The JavaScript 3D library used for the race (release r128).
glTF / GLB	A standard file format for 3D models. The models are stored as GLB inside JavaScript files.
localStorage	Storage inside the WebView where the game saves profiles, progress and records.
CSV	Comma-separated values: the spreadsheet file the teacher exports.
SWBST	Somebody – Wanted – But – So – Then, the summarising method the game teaches.
Story parts	The five SWBST answers for a story, used in the race and in Story order.
Main idea	What the whole story is mostly about; one 4-choice question per story.
Session / Level / World	30 levels are grouped as 10 sessions of 3 (easy, average, hard) and 5 worlds of 6.
Teacher Mira	The cartoon teacher who guides the learner with speech bubbles and voice clips.
Patrol / gap meter	The officer and dog that chase the runner, and the 0–100 distance to them.
Answer gate	The row across the 3 lanes; the lane the runner is in at the gate is the answer.
TTS	Text-to-speech. Mira's clips were made with Kokoro TTS; lines without a clip use the phone's voice.
CC0	Creative Commons public domain dedication: free to use with no credit required.
CC BY / CC BY-SA	Creative Commons licenses that require credit (BY), and sharing changes under the same license (SA).
OFL	SIL Open Font License, used by most of the fonts.
PR	Pull request: a proposed change on GitHub, reviewed before it is merged into main.
Signing key	The private key that signs release APKs. Updates must use the same key.

REFERENCE

References

APA 7th edition. Software and asset packs without a release date are cited as n.d.

Apache Software Foundation. (2004). *Apache License, version 2.0*. https://www.apache.org/licenses/LICENSE-2.0

Braille Institute of America. (n.d.). *Atkinson Hyperlegible* [Typeface]. Google Fonts. https://fonts.google.com/specimen/Atkinson+Hyperlegible

Astigmatic. (n.d.). *Luckiest Guy* [Typeface]. Google Fonts. https://fonts.google.com/specimen/Luckiest+Guy

Brandão, M., & Hafontia. (n.d.). *Fredoka* [Typeface]. Google Fonts. https://fonts.google.com/specimen/Fredoka

Creative Commons. (n.d.-a). *CC0 1.0 Universal*. https://creativecommons.org/publicdomain/zero/1.0/

Creative Commons. (n.d.-b). *Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)*. https://creativecommons.org/licenses/by-sa/4.0/

FFmpeg project. (n.d.). *FFmpeg* [Computer software]. https://ffmpeg.org

hexgrad. (n.d.). *Kokoro-82M* [Text-to-speech model]. Hugging Face. https://huggingface.co/hexgrad/Kokoro-82M

Kenney. (n.d.-a). *City kit (commercial)* [3D models]. https://kenney.nl/assets/city-kit-commercial

Kenney. (n.d.-b). *City kit (suburban)* [3D models]. https://kenney.nl/assets/city-kit-suburban

Kenney. (n.d.-c). *Nature kit* [3D models]. https://kenney.nl/assets/nature-kit

Kenney. (n.d.-d). *Train kit* [3D models]. https://kenney.nl/assets/train-kit

Khronos Group. (n.d.). *UnityGLTF* [Computer software]. GitHub. https://github.com/KhronosGroup/UnityGLTF

Macon, J. M., Bewell, D., & Vogt, M. (1991). *Responses to literature: Grades K–8*. International Reading Association.

McCurdy, D. (n.d.). *glTF-Transform* (Version 4.1.1) [Computer software]. https://gltf-transform.dev

Montoreano, J. (n.d.). *Lilita One* [Typeface]. Google Fonts. https://fonts.google.com/specimen/Lilita+One

Quaternius. (n.d.). *Ultimate animated character pack* [3D models]. https://quaternius.com

SIL International. (n.d.-a). *Andika* [Typeface]. Google Fonts. https://fonts.google.com/specimen/Andika

SIL International. (n.d.-b). *SIL Open Font License, version 1.1*. https://openfontlicense.org

Speer, R. (2022). *wordfreq* (Version 3.0.2) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.7199437

three.js authors. (n.d.). *three.js* (Release r128) [Computer software]. GitHub. https://github.com/mrdoob/three.js

Not yet cited

- **Story texts.** [Full references for the 30 stories, to be supplied by the research team]. Several stories are published children's books, for example *The Day the Crayons Quit* and *Owen and Mzee*.
- **Music and UI images, including Teacher Mira.** [Source and license, to be confirmed by the owner].

> The wordfreq version that `words.js` was taken from is not recorded; version 3.0.2 is the current citable release.

APPENDIX A

Assets and credits

Every file the game ships, where it came from, who made it, and its license. Checked against `main` (1.3.6) on 2026-11-05.

A.1 Overview

This page lists every asset inside `app/src/main/assets/www` and the Android launcher icons. Where the source or license is not recorded anywhere (the original `CREDITS.md` was lost with the original project folder), it says **unknown** instead of guessing. One item is marked **likely** where the file itself points to a source but nothing records it.

3D models

7 files · 4.5 MB · Quaternius and Kenney, CC0

Voice clips

320 MP3 · 8.0 MB · made with Kokoro TTS

Music

2 MP3 · 2.2 MB · source unknown

Images

18 files · 0.9 MB · source unknown

Fonts

7 files · 160 KB · Google Fonts, OFL / Apache

Sounds

11 clips inside assets.js · Kenney, CC0

A.2 Gaps to fix before a public release

These are not guesses; each one is visible in the files.

- **Music and all UI images have no recorded source.** The credits screen calls the music "original loops, made for this game" but there is no record of who made it or with what tool. The files carry a C2PA provenance tag from a file-handling tool, which only shows how they were copied, not who made them. The owner should confirm where Mira, the world pictures, the backdrop, stars, icons and both music loops came from.
- **wordfreq (CC BY-SA 4.0) is not credited on the credits screen.** That license requires credit. Suggested credit: "Word list: wordfreq by Robyn Speer (CC BY-SA 4.0)".
- **The credits screen says "Full list in CREDITS.md"**, but that file was lost and is not in the app. It should point to this document, or a new credits file should be shipped.
- **The font license texts are not shipped.** The SIL OFL and Apache 2.0 both ask for the license text to travel with the font files. Add `OFL.txt` and the Apache license next to `fonts/` .
- **Stories have no author credit.** Several are published children's stories (for example "The Day the Crayons Quit" and "Owen and Mzee"). The research team should confirm they may be used and how to credit them.
- **Teacher Mira has no credit line** on the credits screen, and no repo LICENSE says who owns the game code.

A.3 3D models

Models

Table A.1 Models

File	Size	Used in the game	Source	Author	License
`models/boy.js`	906 KB	Max, the boy runner (animations: Run, Jump, Roll, Idle, Victory, Defeat)	quaternius.com · Ultimate Animated Character Pack	Quaternius	CC0 1.0 (public domain) (as stated on the credits screen)
`models/girl.js`	911 KB	Mia, the girl runner (same animations)	Same pack	Quaternius	CC0 1.0 (public domain)
`models/officer.js`	951 KB	The patrol officer who chases the runner (Run, Idle, Victory)	Same pack	Quaternius	CC0 1.0 (public domain)
`models/trains.js`	155 KB	Subway trains on the track (train-electric-subway-a/b/c, wheels, connector)	Kenney Train Kit	Kenney (kenney.nl)	CC0 1.0 (public domain)
`models/city.js`	1001 KB	Buildings and skyscrapers beside the track (Story Town)	Kenney City Kit (Commercial)	Kenney (kenney.nl)	CC0 1.0 (public domain)
`models/suburb.js`	467 KB	Houses, trees, fences and planters (meadow and river worlds)	Kenney City Kit (Suburban)	Kenney (kenney.nl)	CC0 1.0 (public domain)
`models/nature.js`	182 KB	Trees, palms, pines, rocks, bushes, flowers, mushrooms, logs	Kenney Nature Kit	Kenney (kenney.nl)	CC0 1.0 (public domain)
`assets.js` → coin, cloud	45 KB	Coins on the track, clouds in the sky	Kenney, exact kit not recorded. likely Platformer Kit: the shared texture is named “platformer”	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → platformer texture	10 KB	Colour map for the coin and cloud models	Same as above	Kenney (kenney.nl)	CC0 1.0 (public domain)

Characters are recoloured and given cartoon shading in code. Mesh names in the files match the Kenney kits listed (for example `train-electric-subway-a`, `building-skyscraper-a`, `building-type-a`, `tree_palmTall`). All model files were compressed with glTF-Transform 4.1.1 and stored as base64.

A.4 Sound effects

Sounds inside assets.js

Table A.2 Sounds inside assets.js

File	Size	Used in the game	Source	Author	License
`assets.js` → click	1 KB	Most button taps	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → tap	1 KB	Small taps	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → good	5 KB	Right answer	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → bad	5 KB	Wrong answer	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)

File	Size	Used in the game	Source	Author	License
`assets.js` → win	2 KB	Level, race or summary passed	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → coin	3 KB	Coin picked up	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → jump	2 KB	Runner jumps	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → land	1 KB	Runner lands	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → swish	1 KB	Page turn, roll	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → bump	3 KB	Hit a train, caught	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
`assets.js` → hit	7 KB	Hit a barrier	Kenney, exact pack not recorded	Kenney (kenney.nl)	CC0 1.0 (public domain) (file header says Kenney CC0)
beep, pop	—	Countdown and small pops	Made in code with Web Audio (no file)	SummaRace code	—

A.5 Music

Music

Table A.3 Music

File	Size	Used in the game	Source	Author	License
`music/menu.mp3`	1.0 MB	Background music on every screen except the race (about 90 s loop)	unknown The credits screen says “Original loops, made for this game”, but who made them and how is not recorded	unknown	unknown
`music/race.mp3`	1.1 MB	Race music (about 72 s loop, A minor, ~161 bpm)	unknown Same as above	unknown	unknown

> Not in the game: three Kevin MacLeod tracks (Pixel Peeker Polka – faster, Monkeys Spinning Monkeys, Run Amok, from incompetech.com, CC BY 4.0) were downloaded as previews only. If one is used, it needs the credit line "Music by Kevin MacLeod (incompetech.com), licensed under Creative Commons: By Attribution 4.0".

A.6 Voice: Teacher Mira

Table A.4 Voice clips: summary

Item	Detail
Files	`voice/m-000.mp3` to `voice/m-319.mp3` : 320 clips, 8.0 MB. 170 are Mira's lines, 150 are the 30 stories × 5 pages read aloud.
Made with	Kokoro TTS, recorded offline (credits screen and `voice.js` header). Model: hexgrad/Kokoro-82M.
Author	Model by hexgrad. Which Kokoro voice was used: unknown
License	Apache License 2.0 (the model). The clips were generated for this game.
Build script	`build-voice.sh` made the clips and `voice.js` . It was lost, so new lines cannot be recorded in the same voice yet.

Item	Detail
Fallback	A line with no clip is spoken by the phone's own voice (Android TextToSpeech, en-US) or the browser's voice.

Lines that have no clip

All 80 lines in Mira's main line lists have clips. These fixed lines in the code do not, so they use the phone voice:

- Time's up! Here is the right order. Read it once, then let's write your summary.
- Last stage! Write 2 or 3 sentences that tell the story in your own words. Before you check, ask yourself: ...
- Tell me more! Write 2 or 3 sentences: who the story is about, what happened, and how it ended.
- Some words are repeated too many times. Tell the story in sentences that make sense.
- Write it in sentences, not a list of words. Use words like wanted, but, so, and then.
- I see only one sentence. Please write 2 or 3 sentences, with a period at the end of each one.
- Some details don't match the story. Look at the red lines and fix them.
- Who is the story about? Name the main character in your summary.
- Your summary needs the main idea. What is the whole story mostly about?
- Good effort! Your teacher will read your summary and help you with it.

Lines built from a template (with a name, number or story title) were not checked one by one and may also fall back.

▼ **Full list of all 320 voice clips** (line text is stored in lower case without punctuation)

Table A.5 All 320 voice clips

File	Kind	Line / page	Size
`m-000.mp3`	Mira line	hi i m teacher mira type your name so i can save your progress then we ll start reading together	28.8 KB
`m-001.mp3`	Mira line	please type your name first	12.3 KB
`m-002.mp3`	Mira line	that level is still locked finish the open level first and it will unlock	23.1 KB
`m-003.mp3`	Mira line	welcome start with level 1 you will read answer put events in order race the patrol and then summarize	32.1 KB
`m-004.mp3`	Mira line	you finished all 30 stories replay any level to earn more stars	22.1 KB
`m-005.mp3`	Mira line	welcome back your next level is ready for you	16.4 KB
`m-006.mp3`	Mira line	read carefully look for somebody what they wanted the problem what they did and how it ended	28.1 KB
`m-007.mp3`	Mira line	take your time think who is the story about and what is the big idea	23.1 KB
`m-008.mp3`	Mira line	picture the story in your head it helps you remember what happened first next then and last	27.1 KB
`m-009.mp3`	Mira line	let s check the key details choose the best answer	17.7 KB
`m-010.mp3`	Mira line	now the big one the main idea which answer tells what the whole story is about not just one small part	32.2 KB
`m-011.mp3`	Mira line	correct great memory	12.0 KB
`m-012.mp3`	Mira line	yes that s right	10.6 KB

File	Kind	Line / page	Size
m-013.mp3	Mira line	exactly you read closely	13.2 KB
m-014.mp3	Mira line	not quite the main idea covers the whole story look at the green answer	23.5 KB
m-015.mp3	Mira line	not quite look at the green answer and the reason below	19.4 KB
m-016.mp3	Mira line	put the story in order tap the event that happened first then next then then and last tap a placed event to take it back	35.6 KB
m-017.mp3	Mira line	perfect order on the first try now the race pick the lanes with the right answers to escape the patrol	31.7 KB
m-018.mp3	Mira line	all in order now time for the race pick the right answer lanes to escape	23.7 KB
m-019.mp3	Mira line	you escaped now find the story elements choose the best answer for somebody wanted but so and then	31.3 KB
m-020.mp3	Mira line	all five story elements are correct now use them to write your summary	22.8 KB
m-021.mp3	Mira line	now all five are correct let s use them to write your summary	20.3 KB
m-022.mp3	Mira line	last stage join the story elements into one sentence using your own words use words like wanted but so and then	36.0 KB
m-023.mp3	Mira line	i see more than one sentence join your ideas into one sentence with words like but so and then	30.4 KB
m-024.mp3	Mira line	that sentence is very long a summary keeps only the main idea try to use fewer words	28.7 KB
m-025.mp3	Mira line	part of this is copied straight from the story try saying it in your own words	23.4 KB
m-026.mp3	Mira line	excellent your sentence has every story element	17.1 KB
m-027.mp3	Mira line	the red chips show what is still missing add them in your own words	20.3 KB
m-028.mp3	Mira line	you completed the final story i m so proud of you	17.8 KB
m-029.mp3	Mira line	replay this level anytime to earn more stars	17.8 KB
m-030.mp3	Mira line	three stars you re a true summaracer	14.4 KB
m-031.mp3	Mira line	the next level is now unlocked replay this level anytime to earn more stars	25.2 KB
m-032.mp3	Mira line	the next level is now unlocked three stars you re a true summaracer	23.2 KB
m-033.mp3	Mira line	read the question then pick your lane	14.4 KB
m-034.mp3	Mira line	which lane has the right answer	12.8 KB
m-035.mp3	Mira line	think about the story then move	13.5 KB
m-036.mp3	Mira line	watch out for trains	11.2 KB

File	Kind	Line / page	Size
m-037.mp3	Mira line	yes you remembered	11.2 KB
m-038.mp3	Mira line	great answer keep going	12.3 KB
m-039.mp3	Mira line	correct the patrol is falling behind	15.2 KB
m-040.mp3	Mira line	oops that was not the right answer the patrol is closing in	20.0 KB
m-041.mp3	Mira line	trains block the lane switch lanes early	16.5 KB
m-042.mp3	Mira line	jump over the low barriers or switch lanes	15.4 KB
m-043.mp3	Mira line	swipe down to roll under the high barriers	15.3 KB
m-044.mp3	Mira line	the patrol is right behind you pick the correct answer lanes to get away	22.8 KB
m-045.mp3	Mira line	your device can t read aloud but you can read it yourself	19.1 KB
m-046.mp3	Mira line	0 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	35.2 KB
m-047.mp3	Mira line	1 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	34.3 KB
m-048.mp3	Mira line	2 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	34.5 KB
m-049.mp3	Mira line	3 of 4 are in the right place the green ones stay ask yourself what had to happen first try the red ones again	34.4 KB
m-050.mp3	Mira line	0 of 5 are correct change the red ones and check again	20.9 KB
m-051.mp3	Mira line	1 of 5 are correct change the red ones and check again	19.9 KB
m-052.mp3	Mira line	2 of 5 are correct change the red ones and check again	19.9 KB
m-053.mp3	Mira line	3 of 5 are correct change the red ones and check again	19.8 KB
m-054.mp3	Mira line	4 of 5 are correct change the red ones and check again	20.1 KB
m-055.mp3	Mira line	that s only 1 word a good one sentence summary has about 15 to 30 words	26.1 KB
m-056.mp3	Mira line	that s only 2 words a good one sentence summary has about 15 to 30 words	26.2 KB
m-057.mp3	Mira line	that s only 3 words a good one sentence summary has about 15 to 30 words	26.8 KB
m-058.mp3	Mira line	that s only 4 words a good one sentence summary has about 15 to 30 words	26.6 KB
m-059.mp3	Mira line	that s only 5 words a good one sentence summary has about 15 to 30 words	27.0 KB
m-060.mp3	Mira line	that s only 6 words a good one sentence summary has about 15 to 30 words	26.5 KB

File	Kind	Line / page	Size
m-061.mp3	Mira line	that s only 7 words a good one sentence summary has about 15 to 30 words	27.1 KB
m-062.mp3	Mira line	that s only 8 words a good one sentence summary has about 15 to 30 words	26.2 KB
m-063.mp3	Mira line	that s only 9 words a good one sentence summary has about 15 to 30 words	26.9 KB
m-064.mp3	Mira line	good summary you included 3 of 5 story elements	18.6 KB
m-065.mp3	Mira line	good summary you included 4 of 5 story elements	18.7 KB
m-066.mp3	Mira line	you included 0 of 5 story elements look at the story elements above and add the missing parts	30.6 KB
m-067.mp3	Mira line	you included 1 of 5 story elements look at the story elements above and add the missing parts	29.3 KB
m-068.mp3	Mira line	you included 2 of 5 story elements look at the story elements above and add the missing parts	29.8 KB
m-069.mp3	Mira line	you included 3 of 5 story elements look at the story elements above and add the missing parts	29.4 KB
m-070.mp3	Mira line	you included 4 of 5 story elements look at the story elements above and add the missing parts	29.9 KB
m-071.mp3	Mira line	hi i m teacher mira tap play then type your name so i can save your progress	25.8 KB
m-072.mp3	Mira line	type your name or tap your name if you played before	17.4 KB
m-073.mp3	Mira line	let s read together who is this story about what do they want	19.9 KB
m-074.mp3	Mira line	what problem happens keep reading carefully	16.5 KB
m-075.mp3	Mira line	how does the story end remember what happened first next then and last	22.9 KB
m-076.mp3	Mira line	here is the story again read the yellow part carefully then try again	23.1 KB
m-077.mp3	Mira line	here is the story again read carefully then try again	20.0 KB
m-078.mp3	Mira line	not quite do you want to read the story again	15.8 KB
m-079.mp3	Mira line	choose another answer you can do it	14.3 KB
m-080.mp3	Mira line	some events are not in the right place yet the green ones stay do you want to read the story again	27.9 KB
m-081.mp3	Mira line	ask yourself what had to happen first	15.2 KB
m-082.mp3	Mira line	you escaped now find the story elements one at a time	19.6 KB
m-083.mp3	Mira line	i can t read some of these words please write your summary in english using real words	26.3 KB
m-084.mp3	Mira line	some words are repeated too many times tell the story in a sentence that makes sense	25.3 KB

File	Kind	Line / page	Size
m-085.mp3	Mira line	good summary you told the main idea in your own words	18.1 KB
m-086.mp3	Mira line	who is the story about name the main character in your sentence	20.5 KB
m-087.mp3	Mira line	your sentence is missing some story elements look at the story elements above and add the missing parts	30.7 KB
m-088.mp3	Mira line	your sentence needs the main idea what is the whole story mostly about	24.1 KB
m-089.mp3	Mira line	which card has the right answer	12.7 KB
m-090.mp3	Mira line	this question is back think about the story and try again	19.3 KB
m-091.mp3	Mira line	not quite that question will come back the patrol is closing in	20.9 KB
m-092.mp3	Mira line	you got them all run to the finish line	15.2 KB
m-093.mp3	Mira line	you made it mission complete	13.1 KB
m-094.mp3	Mira line	write it as a sentence not a list of words use words like wanted but so and then	26.7 KB
m-095.mp3	Mira line	you got it	9.4 KB
m-096.mp3	Mira line	that s it well done	11.1 KB
m-097.mp3	Mira line	nice work that s correct	12.1 KB
m-098.mp3	Mira line	super you remembered	11.6 KB
m-099.mp3	Mira line	right answer keep it up	12.3 KB
m-100.mp3	Mira line	wonderful you were paying attention	14.3 KB
m-101.mp3	Mira line	yes you found it in the story	12.9 KB
m-102.mp3	Mira line	great thinking	10.1 KB
m-103.mp3	Mira line	perfect on to the next one	12.5 KB
m-104.mp3	Mira line	awesome that s the one	11.9 KB
m-105.mp3	Mira line	way to go	9.5 KB
m-106.mp3	Mira line	you re a careful reader	11.0 KB
m-107.mp3	Mira line	good job that s correct	12.0 KB
m-108.mp3	Mira line	brilliant you know this story	12.4 KB

File	Kind	Line / page	Size
m-109.mp3	Mira line	yes indeed nicely done	12.9 KB
m-110.mp3	Mira line	hmm not that one want to look at the story again	17.3 KB
m-111.mp3	Mira line	almost let s check the story together	14.8 KB
m-112.mp3	Mira line	not yet the story can help you want to read it again	18.5 KB
m-113.mp3	Mira line	good try let s look back at the story	15.4 KB
m-114.mp3	Mira line	oops not that one shall we read that part again	17.0 KB
m-115.mp3	Mira line	that s not it yet take another look at the story	16.9 KB
m-116.mp3	Mira line	close but not quite want to peek at the story	16.9 KB
m-117.mp3	Mira line	try again think about what you read	14.5 KB
m-118.mp3	Mira line	take your time and try another one	14.1 KB
m-119.mp3	Mira line	you can do this pick again	12.7 KB
m-120.mp3	Mira line	read the question once more then choose	15.2 KB
m-121.mp3	Mira line	think about the story then try again	14.6 KB
m-122.mp3	Mira line	don t give up try a different answer	14.2 KB
m-123.mp3	Mira line	let s turn the page	11.0 KB
m-124.mp3	Mira line	here s the next page	11.1 KB
m-125.mp3	Mira line	on to the next page	11.1 KB
m-126.mp3	Mira line	let s keep reading	10.8 KB
m-127.mp3	Mira line	ready for more here s the next page	15.0 KB
m-128.mp3	Mira line	now a question about this page	13.2 KB
m-129.mp3	Mira line	let s see what you remember from this page	14.2 KB
m-130.mp3	Mira line	question time think about what you just read	16.4 KB
m-131.mp3	Mira line	here s a question about that page	13.5 KB
m-132.mp3	Mira line	here s a new story read carefully and look for the main character	20.6 KB

File	Kind	Line / page	Size
m-133.mp3	Mira line	story time who is in this story and what do they want	18.6 KB
m-134.mp3	Mira line	let s begin read slowly and picture the story in your mind	20.2 KB
m-135.mp3	Mira line	keep going watch for the problem in the story	16.4 KB
m-136.mp3	Mira line	what is happening now read carefully	14.7 KB
m-137.mp3	Mira line	think about what the character does next	14.1 KB
m-138.mp3	Mira line	which card has the right story part	14.0 KB
m-139.mp3	Mira line	find the right card and run into it	14.4 KB
m-140.mp3	Mira line	look at all three cards then choose	14.5 KB
m-141.mp3	Mira line	remember the story which one fits	14.5 KB
m-142.mp3	Mira line	boost that s the right part	12.6 KB
m-143.mp3	Mira line	nice catch keep running	12.1 KB
m-144.mp3	Mira line	you got it speed up	11.3 KB
m-145.mp3	Mira line	perfect the patrol can t keep up	14.4 KB
m-146.mp3	Mira line	not quite that one will come back the patrol is closing in	19.3 KB
m-147.mp3	Mira line	oops wrong card you ll see it again soon	15.9 KB
m-148.mp3	Mira line	not that one think about the story and try again later	19.0 KB
m-149.mp3	Mira line	careful that part will come back run faster	17.1 KB
m-150.mp3	Mira line	this one is back think about the story and try again	18.4 KB
m-151.mp3	Mira line	here it is again which card is right this time	16.8 KB
m-152.mp3	Mira line	second chance remember what happened in the story	17.0 KB
m-153.mp3	Mira line	all the story parts are yours race to the finish	17.2 KB
m-154.mp3	Mira line	that s all five now run to the finish line	16.2 KB
m-155.mp3	Mira line	finish line you escaped the patrol	14.2 KB
m-156.mp3	Mira line	hooray mission complete	12.2 KB

File	Kind	Line / page	Size
m-157.mp3	Mira line	this is the last page how does it all end	15.7 KB
m-158.mp3	Mira line	last page think about how the problem was solved	17.1 KB
m-159.mp3	Mira line	almost done read how the story ends	15.3 KB
m-160.mp3	Mira line	great running now put the story parts in order tap the somebody part first then wanted but so and then tap a placed part to take it back	41.4 KB
m-161.mp3	Mira line	perfect order on the first try now use these parts to write your summary	23.1 KB
m-162.mp3	Mira line	all in order now now use these parts to write your summary	20.0 KB
m-163.mp3	Mira line	some parts are not in the right place yet the green ones stay do you want to read the story again	27.6 KB
m-164.mp3	Mira line	ask yourself who is it about what did they want and what went wrong	20.6 KB
m-165.mp3	Mira line	the race could not start on this device so let s keep going	18.9 KB
m-166.mp3	Mira line	last stage write one sentence that tells the main idea of the story in your own words before you check ask yourself did i tell who what they wanted the problem what they did and how it ended	55.2 KB
m-167.mp3	Mira line	your sentence is missing some story parts did you tell who what they wanted the problem what they did and how it ended	34.6 KB
m-168.mp3	Mira line	the patrol is right behind you collect the right story parts to get away	23.1 KB
m-169.mp3	Mira line	welcome start with session 1 you will read and answer race the patrol to collect the story parts put them in order and then write a summary	39.8 KB
page-01-1.mp3	Story page	Level 1 “The Playground”, page 1	47.8 KB
page-01-2.mp3	Story page	Level 1 “The Playground”, page 2	44.1 KB
page-01-3.mp3	Story page	Level 1 “The Playground”, page 3	57.2 KB
page-01-4.mp3	Story page	Level 1 “The Playground”, page 4	73.2 KB
page-01-5.mp3	Story page	Level 1 “The Playground”, page 5	29.6 KB
page-02-1.mp3	Story page	Level 2 “The Day the Crayons Quit”, page 1	35.3 KB
page-02-2.mp3	Story page	Level 2 “The Day the Crayons Quit”, page 2	18.0 KB
page-02-3.mp3	Story page	Level 2 “The Day the Crayons Quit”, page 3	23.8 KB
page-02-4.mp3	Story page	Level 2 “The Day the Crayons Quit”, page 4	30.4 KB
page-02-5.mp3	Story page	Level 2 “The Day the Crayons Quit”, page 5	24.3 KB
page-03-1.mp3	Story page	Level 3 “The Animal Assignment”, page 1	29.9 KB

File	Kind	Line / page	Size
page-03-2.mp3	Story page	Level 3 “The Animal Assignment”, page 2	29.5 KB
page-03-3.mp3	Story page	Level 3 “The Animal Assignment”, page 3	24.6 KB
page-03-4.mp3	Story page	Level 3 “The Animal Assignment”, page 4	29.5 KB
page-03-5.mp3	Story page	Level 3 “The Animal Assignment”, page 5	34.3 KB
page-04-1.mp3	Story page	Level 4 “Escaping the Room”, page 1	24.8 KB
page-04-2.mp3	Story page	Level 4 “Escaping the Room”, page 2	22.0 KB
page-04-3.mp3	Story page	Level 4 “Escaping the Room”, page 3	19.4 KB
page-04-4.mp3	Story page	Level 4 “Escaping the Room”, page 4	27.1 KB
page-04-5.mp3	Story page	Level 4 “Escaping the Room”, page 5	24.5 KB
page-05-1.mp3	Story page	Level 5 “Seeking a Friend”, page 1	31.4 KB
page-05-2.mp3	Story page	Level 5 “Seeking a Friend”, page 2	19.7 KB
page-05-3.mp3	Story page	Level 5 “Seeking a Friend”, page 3	16.3 KB
page-05-4.mp3	Story page	Level 5 “Seeking a Friend”, page 4	22.0 KB
page-05-5.mp3	Story page	Level 5 “Seeking a Friend”, page 5	25.9 KB
page-06-1.mp3	Story page	Level 6 “Shawn the Speedy Snail”, page 1	36.2 KB
page-06-2.mp3	Story page	Level 6 “Shawn the Speedy Snail”, page 2	22.8 KB
page-06-3.mp3	Story page	Level 6 “Shawn the Speedy Snail”, page 3	30.2 KB
page-06-4.mp3	Story page	Level 6 “Shawn the Speedy Snail”, page 4	31.0 KB
page-06-5.mp3	Story page	Level 6 “Shawn the Speedy Snail”, page 5	30.9 KB
page-07-1.mp3	Story page	Level 7 “Emma's Favorite Restaurant”, page 1	32.6 KB
page-07-2.mp3	Story page	Level 7 “Emma's Favorite Restaurant”, page 2	22.0 KB
page-07-3.mp3	Story page	Level 7 “Emma's Favorite Restaurant”, page 3	21.5 KB
page-07-4.mp3	Story page	Level 7 “Emma's Favorite Restaurant”, page 4	21.8 KB
page-07-5.mp3	Story page	Level 7 “Emma's Favorite Restaurant”, page 5	25.9 KB

File	Kind	Line / page	Size
page-08-1.mp3	Story page	Level 8 “The Case of the Missing Lunch”, page 1	27.4 KB
page-08-2.mp3	Story page	Level 8 “The Case of the Missing Lunch”, page 2	19.9 KB
page-08-3.mp3	Story page	Level 8 “The Case of the Missing Lunch”, page 3	20.8 KB
page-08-4.mp3	Story page	Level 8 “The Case of the Missing Lunch”, page 4	26.4 KB
page-08-5.mp3	Story page	Level 8 “The Case of the Missing Lunch”, page 5	24.8 KB
page-09-1.mp3	Story page	Level 9 “The Selfish Giant”, page 1	27.6 KB
page-09-2.mp3	Story page	Level 9 “The Selfish Giant”, page 2	22.3 KB
page-09-3.mp3	Story page	Level 9 “The Selfish Giant”, page 3	20.1 KB
page-09-4.mp3	Story page	Level 9 “The Selfish Giant”, page 4	23.5 KB
page-09-5.mp3	Story page	Level 9 “The Selfish Giant”, page 5	17.8 KB
page-10-1.mp3	Story page	Level 10 “A Visit to the Zoo”, page 1	27.6 KB
page-10-2.mp3	Story page	Level 10 “A Visit to the Zoo”, page 2	14.5 KB
page-10-3.mp3	Story page	Level 10 “A Visit to the Zoo”, page 3	22.1 KB
page-10-4.mp3	Story page	Level 10 “A Visit to the Zoo”, page 4	17.2 KB
page-10-5.mp3	Story page	Level 10 “A Visit to the Zoo”, page 5	32.5 KB
page-11-1.mp3	Story page	Level 11 “Be Careful What You Wish For”, page 1	35.4 KB
page-11-2.mp3	Story page	Level 11 “Be Careful What You Wish For”, page 2	29.6 KB
page-11-3.mp3	Story page	Level 11 “Be Careful What You Wish For”, page 3	36.1 KB
page-11-4.mp3	Story page	Level 11 “Be Careful What You Wish For”, page 4	26.3 KB
page-11-5.mp3	Story page	Level 11 “Be Careful What You Wish For”, page 5	33.7 KB
page-12-1.mp3	Story page	Level 12 “AgitAgueda”, page 1	35.8 KB
page-12-2.mp3	Story page	Level 12 “AgitAgueda”, page 2	26.5 KB
page-12-3.mp3	Story page	Level 12 “AgitAgueda”, page 3	29.5 KB
page-12-4.mp3	Story page	Level 12 “AgitAgueda”, page 4	36.6 KB

File	Kind	Line / page	Size
page-12-5.mp3	Story page	Level 12 “AgitAgueda”, page 5	28.6 KB
page-13-1.mp3	Story page	Level 13 “The Crowded House: A Folktale”, page 1	38.1 KB
page-13-2.mp3	Story page	Level 13 “The Crowded House: A Folktale”, page 2	23.8 KB
page-13-3.mp3	Story page	Level 13 “The Crowded House: A Folktale”, page 3	34.7 KB
page-13-4.mp3	Story page	Level 13 “The Crowded House: A Folktale”, page 4	35.5 KB
page-13-5.mp3	Story page	Level 13 “The Crowded House: A Folktale”, page 5	32.4 KB
page-14-1.mp3	Story page	Level 14 “Owen and Mzee”, page 1	32.9 KB
page-14-2.mp3	Story page	Level 14 “Owen and Mzee”, page 2	20.6 KB
page-14-3.mp3	Story page	Level 14 “Owen and Mzee”, page 3	27.6 KB
page-14-4.mp3	Story page	Level 14 “Owen and Mzee”, page 4	20.7 KB
page-14-5.mp3	Story page	Level 14 “Owen and Mzee”, page 5	23.4 KB
page-15-1.mp3	Story page	Level 15 “In Grandfather's Day”, page 1	29.1 KB
page-15-2.mp3	Story page	Level 15 “In Grandfather's Day”, page 2	23.8 KB
page-15-3.mp3	Story page	Level 15 “In Grandfather's Day”, page 3	26.1 KB
page-15-4.mp3	Story page	Level 15 “In Grandfather's Day”, page 4	27.0 KB
page-15-5.mp3	Story page	Level 15 “In Grandfather's Day”, page 5	30.5 KB
page-16-1.mp3	Story page	Level 16 “You Can't Always Tell”, page 1	35.3 KB
page-16-2.mp3	Story page	Level 16 “You Can't Always Tell”, page 2	31.8 KB
page-16-3.mp3	Story page	Level 16 “You Can't Always Tell”, page 3	29.7 KB
page-16-4.mp3	Story page	Level 16 “You Can't Always Tell”, page 4	30.7 KB
page-16-5.mp3	Story page	Level 16 “You Can't Always Tell”, page 5	37.2 KB
page-17-1.mp3	Story page	Level 17 “Shuffle to Buffalo”, page 1	26.1 KB
page-17-2.mp3	Story page	Level 17 “Shuffle to Buffalo”, page 2	20.8 KB
page-17-3.mp3	Story page	Level 17 “Shuffle to Buffalo”, page 3	24.6 KB

File	Kind	Line / page	Size
page-17-4.mp3	Story page	Level 17 “Shuffle to Buffalo”, page 4	26.8 KB
page-17-5.mp3	Story page	Level 17 “Shuffle to Buffalo”, page 5	25.3 KB
page-18-1.mp3	Story page	Level 18 “In Grandfather's Day”, page 1	27.0 KB
page-18-2.mp3	Story page	Level 18 “In Grandfather's Day”, page 2	27.1 KB
page-18-3.mp3	Story page	Level 18 “In Grandfather's Day”, page 3	32.3 KB
page-18-4.mp3	Story page	Level 18 “In Grandfather's Day”, page 4	22.1 KB
page-18-5.mp3	Story page	Level 18 “In Grandfather's Day”, page 5	32.9 KB
page-19-1.mp3	Story page	Level 19 “The Boy With the Ball”, page 1	28.9 KB
page-19-2.mp3	Story page	Level 19 “The Boy With the Ball”, page 2	15.6 KB
page-19-3.mp3	Story page	Level 19 “The Boy With the Ball”, page 3	23.7 KB
page-19-4.mp3	Story page	Level 19 “The Boy With the Ball”, page 4	23.5 KB
page-19-5.mp3	Story page	Level 19 “The Boy With the Ball”, page 5	21.8 KB
page-20-1.mp3	Story page	Level 20 “Sploosh!”, page 1	27.9 KB
page-20-2.mp3	Story page	Level 20 “Sploosh!”, page 2	21.6 KB
page-20-3.mp3	Story page	Level 20 “Sploosh!”, page 3	27.5 KB
page-20-4.mp3	Story page	Level 20 “Sploosh!”, page 4	26.1 KB
page-20-5.mp3	Story page	Level 20 “Sploosh!”, page 5	34.9 KB
page-21-1.mp3	Story page	Level 21 “First Day at the Factory”, page 1	26.7 KB
page-21-2.mp3	Story page	Level 21 “First Day at the Factory”, page 2	22.7 KB
page-21-3.mp3	Story page	Level 21 “First Day at the Factory”, page 3	25.2 KB
page-21-4.mp3	Story page	Level 21 “First Day at the Factory”, page 4	25.5 KB
page-21-5.mp3	Story page	Level 21 “First Day at the Factory”, page 5	29.0 KB
page-22-1.mp3	Story page	Level 22 “An Ice Idea”, page 1	30.0 KB
page-22-2.mp3	Story page	Level 22 “An Ice Idea”, page 2	23.4 KB

File	Kind	Line / page	Size
page-22-3.mp3	Story page	Level 22 “An Ice Idea”, page 3	19.6 KB
page-22-4.mp3	Story page	Level 22 “An Ice Idea”, page 4	29.5 KB
page-22-5.mp3	Story page	Level 22 “An Ice Idea”, page 5	32.0 KB
page-23-1.mp3	Story page	Level 23 “How to Skateboard”, page 1	25.0 KB
page-23-2.mp3	Story page	Level 23 “How to Skateboard”, page 2	23.8 KB
page-23-3.mp3	Story page	Level 23 “How to Skateboard”, page 3	22.5 KB
page-23-4.mp3	Story page	Level 23 “How to Skateboard”, page 4	26.2 KB
page-23-5.mp3	Story page	Level 23 “How to Skateboard”, page 5	27.1 KB
page-24-1.mp3	Story page	Level 24 “Anansi and the Cook Pots”, page 1	28.8 KB
page-24-2.mp3	Story page	Level 24 “Anansi and the Cook Pots”, page 2	20.3 KB
page-24-3.mp3	Story page	Level 24 “Anansi and the Cook Pots”, page 3	31.4 KB
page-24-4.mp3	Story page	Level 24 “Anansi and the Cook Pots”, page 4	25.1 KB
page-24-5.mp3	Story page	Level 24 “Anansi and the Cook Pots”, page 5	29.1 KB
page-25-1.mp3	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 1	32.2 KB
page-25-2.mp3	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 2	21.4 KB
page-25-3.mp3	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 3	20.0 KB
page-25-4.mp3	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 4	25.8 KB
page-25-5.mp3	Story page	Level 25 “A Pool Fit for a Hedgehog”, page 5	23.7 KB
page-26-1.mp3	Story page	Level 26 “First Fast”, page 1	24.1 KB
page-26-2.mp3	Story page	Level 26 “First Fast”, page 2	21.4 KB
page-26-3.mp3	Story page	Level 26 “First Fast”, page 3	25.6 KB
page-26-4.mp3	Story page	Level 26 “First Fast”, page 4	27.2 KB
page-26-5.mp3	Story page	Level 26 “First Fast”, page 5	34.7 KB
page-27-1.mp3	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 1	33.9 KB

File	Kind	Line / page	Size
page-27-2.mp3	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 2	21.3 KB
page-27-3.mp3	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 3	23.0 KB
page-27-4.mp3	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 4	28.6 KB
page-27-5.mp3	Story page	Level 27 “Baba Yaga, the Girl, and the Hedgehog”, page 5	28.3 KB
page-28-1.mp3	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 1	31.1 KB
page-28-2.mp3	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 2	23.9 KB
page-28-3.mp3	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 3	36.5 KB
page-28-4.mp3	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 4	33.9 KB
page-28-5.mp3	Story page	Level 28 “Why Does the Ocean Have Waves?”, page 5	32.6 KB
page-29-1.mp3	Story page	Level 29 “Clara Barton: Civil War Hero”, page 1	31.1 KB
page-29-2.mp3	Story page	Level 29 “Clara Barton: Civil War Hero”, page 2	24.8 KB
page-29-3.mp3	Story page	Level 29 “Clara Barton: Civil War Hero”, page 3	24.6 KB
page-29-4.mp3	Story page	Level 29 “Clara Barton: Civil War Hero”, page 4	28.4 KB
page-29-5.mp3	Story page	Level 29 “Clara Barton: Civil War Hero”, page 5	36.6 KB
page-30-1.mp3	Story page	Level 30 “We Also Serve”, page 1	33.9 KB
page-30-2.mp3	Story page	Level 30 “We Also Serve”, page 2	25.2 KB
page-30-3.mp3	Story page	Level 30 “We Also Serve”, page 3	34.9 KB
page-30-4.mp3	Story page	Level 30 “We Also Serve”, page 4	32.0 KB
page-30-5.mp3	Story page	Level 30 “We Also Serve”, page 5	31.9 KB

A.7 Fonts

Fonts

Table A.6 Fonts

File	Size	Used in the game	Source	Author	License
`fonts/LuckiestGuy-Regular.woff2`	26 KB	Logo, big titles, scores	Google Fonts: Luckiest Guy	Astigmatic	Apache License 2.0
`fonts/Fredoka-Variable.woff2`	28 KB	Buttons and interface text	Google Fonts: Fredoka	Milena Brandão, Hafontia	SIL Open Font License 1.1
`fonts/Andika-Regular.woff2`	19 KB	Story text, questions, answers	Google Fonts: Andika	SIL International	SIL Open Font License 1.1
`fonts/Andika-Bold.woff2`	19 KB	Bold story text	Same	SIL International	SIL Open Font License 1.1
`fonts/LilitaOne-Regular.woff2`	12 KB	Back-up interface font	Google Fonts: Lilita One	Juan Montoreano	SIL Open Font License 1.1
`fonts/AtkinsonHyperlegible-Regular.woff2`	23 KB	Back-up reading font	Google Fonts: Atkinson Hyperlegible	Braille Institute of America	SIL Open Font License 1.1
`fonts/AtkinsonHyperlegible-Bold.woff2`	23 KB	Back-up reading font, bold	Same	Braille Institute of America	SIL Open Font License 1.1

All fonts are bundled in the app; nothing is loaded from Google at run time.

A.8 Images

Images

Table A.7 Images

File	Size	Used in the game	Source	Author	License
`ui/mira-hello.png`	29 KB	Teacher Mira waving on the title screen	unknown	unknown	unknown
`ui/mira-talk.png`	16 KB	Mira talking (default mood in the dock)	unknown	unknown	unknown
`ui/mira-happy.png`	20 KB	Mira after a right answer	unknown	unknown	unknown
`ui/mira-think.png`	17 KB	Mira after a wrong answer	unknown	unknown	unknown
`ui/mira-wave.png`	17 KB	Mira greeting	unknown	unknown	unknown
`ui/mira-cheer.png`	14 KB	Mira on the results screen	unknown	unknown	unknown
`ui/mira.png`	15 KB	Mira's small face in the race speech bubble	unknown	unknown	unknown
`ui/mira-full.png`	23 KB	**Not used** by the current code	unknown	unknown	unknown
`ui/backdrop.jpg`	164 KB	Title screen and general background	unknown	unknown	unknown

File	Size	Used in the game	Source	Author	License
`ui/world-0.jpg`	111 KB	Sunny Meadow page art and backdrop	unknown	unknown	unknown
`ui/world-1.jpg`	114 KB	River Bend	unknown	unknown	unknown
`ui/world-2.jpg`	137 KB	Story Town	unknown	unknown	unknown
`ui/world-3.jpg`	113 KB	Palm Beach	unknown	unknown	unknown
`ui/world-4.jpg`	106 KB	Summit Peak	unknown	unknown	unknown
`ui/star.png`	7 KB	Earned star	unknown	unknown	unknown
`ui/star-off.png`	7 KB	Empty star	unknown	unknown	unknown
`icons/icon-192.png`	16 KB	Browser tab icon	unknown	unknown	unknown
`icons/icon-512.png`	37 KB	Large app icon	unknown	unknown	unknown
`res/mipmap-*/ic_launcher*.png`	—	Android launcher icon (all sizes)	unknown	unknown	unknown
Mira's small SVG face, map, coins, buttons, sky, scenery made in code	—	Drawn in HTML, CSS, SVG and three.js code	SummaRace code	SummaRace developer(s)	Part of the game

A.9 Code libraries and data

Code and data

Table A.8 Code and data

File	Size	Used in the game	Source	Author	License
`three.min.js`	589 KB	3D engine for the race	github.com/mrdoob/three.js (release r128)	three.js authors	MIT
`GLTFLoader.js`	94 KB	Loads the GLB models	three.js examples (`examples/js/loaders/GLTFLoader.js`), exact release not recorded	three.js authors	MIT
`words.js`	195 KB	24,887 common English words for the summary spelling check	wordfreq word list	Robyn Speer	CC BY-SA 4.0 (data); the file is shared under the same license
`stories.js`	174 KB	The 30 stories, questions and story parts	Research team's study materials. Wrong choices, race wording and model summaries were written for the game	Research team (client); game additions by the developer(s)	Not stated. Original story authors are not listed in the file
`voice/voice.js`	18 KB	Index from Mira's lines to voice clips	Generated by `build-voice.sh` (lost)	SummaRace developer(s)	Part of the game
`index.html`	288 KB	The whole game: screens, rules, race	Written for SummaRace	SummaRace developer(s)	Not stated (no LICENSE file in the repo)

The Android app (`MainActivity.java`) uses only the Android SDK. There are no third-party Android libraries.

A.10 Tools used to make assets

Table A.9 Tools used to make assets

Tool	Used for	Link
glTF-Transform 4.1.1	Compressing the GLB models (stamped in each file)	gltf-transform.dev
UnityGLTF	Exporter stamped inside the coin and cloud files	github.com/KhronosGroup/UnityGLTF
Kokoro TTS	Mira's voice clips	huggingface.co/hexgrad/Kokoro-82M
FFmpeg (Lavf 60.16)	Stamped as the encoder in the music files	ffmpeg.org

APPENDIX B

Answer Key

All 30 stories · built from `stories.js` on `main` (version 1.3.6)

B.1 How to read this key

The game shuffles the answer choices every time, so the letter (A, B, C) changes from play to play. This key gives the **answer text**, not a letter.

- **Page questions:** one per page, 3 choices. The right answer is in bold; the other two choices are listed after it.
- **Main idea:** 4 choices, the right one is given.
- **Story parts (SWBST):** the right answers are the same for the race (pick the lane) and for Story order (put them Somebody → Then). The 2 wrong race choices are listed after each.
- **Model summary:** Teacher Mira's example, shown after the 3rd summary try. Any summary in the learner's own words that tells these parts passes; it does not have to match.

> This key is in a public repository, by the owner's choice. Learners who find it could look up the answers.

B.2 Contents

Level	Session	Difficulty	Title
1	1	Easy	The Playground
2	1	Average	The Day the Crayons Quit
3	1	Hard	The Animal Assignment
4	2	Easy	Escaping the Room
5	2	Average	Seeking a Friend
6	2	Hard	Shawn the Speedy Snail
7	3	Easy	Emma's Favorite Restaurant
8	3	Average	The Case of the Missing Lunch
9	3	Hard	The Selfish Giant
10	4	Easy	A Visit to the Zoo
11	4	Average	Be Careful What You Wish For
12	4	Hard	AgitAgueda
13	5	Easy	The Crowded House: A Folktale
14	5	Average	Owen and Mzee
15	5	Hard	In Grandfather's Day
16	6	Easy	You Can't Always Tell
17	6	Average	Shuffle to Buffalo
18	6	Hard	In Grandfather's Day
19	7	Easy	The Boy With the Ball

Level	Session	Difficulty	Title
20	7	Average	Sploosh!
21	7	Hard	First Day at the Factory
22	8	Easy	An Ice Idea
23	8	Average	How to Skateboard
24	8	Hard	Anansi and the Cook Pots
25	9	Easy	A Pool Fit for a Hedgehog
26	9	Average	First Fast
27	9	Hard	Baba Yaga, the Girl, and the Hedgehog
28	10	Easy	Why Does the Ocean Have Waves?
29	10	Average	Clara Barton: Civil War Hero
30	10	Hard	We Also Serve

1 The Playground

Level 1 · Session 1 · Easy · Sunny Meadow

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Molly**	Bella · The teacher
2	What did Molly want?	**A turn on the swing**	A new toy · To go home
3	What problem did Molly have?	**Bella would not give her a turn**	She lost her ball · She got hurt on the slide
4	What did Molly do to solve her problem?	**She used an "I message"**	She told the teacher · She pushed Bella
5	What happened in the end?	**Bella got off the swing and Molly was happy**	Molly went home sad · Bella and Molly stopped being friends

Main idea

What is the story mostly about?

Molly solved a problem with her friend Bella at the playground by using an "I message" to express her feelings so they could properly take turns on the swing.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Molly**	Bella · The teacher
Wanted	**a turn on the swing**	to go down the slide · to go home early
But	**Bella would not get off the swing**	the swing was broken that day · Molly forgot how to count
So	**Molly used an "I message" to tell Bella how she felt**	Molly ran to tell the teacher · Molly pushed Bella off the swing
Then	**Bella got off the swing and Molly was happy**	Molly went home feeling sad · Bella and Molly stopped playing

Model summary

Molly wanted her turn on the swing, but Bella would not get off. So Molly told Bella how she felt with an I message. Then Bella got off and Molly was happy.

2 The Day the Crayons Quit

Level 2 · Session 1 · Average · Sunny Meadow

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Duncan**	Beige Crayon · Blue Crayon
2	What did Duncan want to do?	**Color with his crayons**	Throw away the crayons · Buy new crayons
3	What problem did the crayons have?	**They were unhappy with their jobs**	They were broken · They were lost
4	What did Duncan learn from the letters?	**The crayons had different complaints**	The crayons wanted new boxes · The crayons wanted to leave school
5	What happened at the end?	**Duncan tried to help the crayons so they could color again**	Duncan threw away the crayons · The crayons disappeared

Main idea

What is the story mostly about?

Different people have different needs, and listening to them can help solve problems.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Duncan**	Beige Crayon · Blue Crayon
Wanted	**to color with his crayons**	to buy a new box of crayons · to write letters to his friends
But	**the crayons quit because they were unhappy**	the crayons were all broken · his crayon box was lost
So	**Duncan listened to their complaints**	Duncan threw the crayons away · Duncan bought new crayons
Then	**he tried to find a way to make them happy again**	he stopped coloring forever · he gave the crayons to a friend

Model summary

Duncan wanted to draw with his crayons, but they quit because they were unhappy. So he read their letters and listened. Then he tried to make every crayon happy again.

3 The Animal Assignment

Level 3 · Session 1 · Hard · Sunny Meadow

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Jayden**	Mr. Trevino · Landon
2	What did Jayden need to do?	**Pick an animal for his assignment**	Visit a zoo · Draw an animal
3	What problem did Jayden face?	**He could not decide which animal to choose**	He forgot the assignment · He lost his notebook
4	What did Jayden do to solve his problem?	**He kept researching and thinking**	He asked someone else to choose · He ignored the assignment
5	What happened in the end?	**Jayden chose to be a human**	Jayden chose to be a dolphin · Jayden did not finish the assignment

Main idea

What is the story mostly about?
Being human is special because people can learn, communicate, and achieve great things.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Jayden**	Mr. Trevino · Landon
Wanted	**to choose an animal for his assignment**	to visit a zoo with his class · to draw a picture of a fox
But	**every animal he researched had disadvantages**	he forgot to do his homework · his teacher was absent that day
So	**he continued researching and thinking carefully**	he asked Landon to choose for him · he gave up on the assignment
Then	**he chose to stay human**	he chose to write about a dolphin · he did not turn in his work

Model summary

Jayden wanted to pick an animal for his class paper, but every animal had something bad about it. So he kept reading and thinking. Then he picked being a human.

4 Escaping the Room

Level 4 · Session 2 · Easy · Sunny Meadow

Page questions

Page	Question	Right answer	Other choices
1	Who are the main characters?	**Maggie, Travis, and Lucy**	Maggie and her teacher · Lucy and a scientist
2	What did the friends want to do?	**Escape the room**	Find a treasure · Build a puzzle
3	What problem did the friends face?	**The door was locked and the key was stuck in ice**	They lost the key · The room was dark
4	What did Lucy suggest?	**Using a metal tray to melt the ice**	Breaking the key · Calling for help
5	What happened at the end?	**They escaped the room**	They stayed inside · They lost the key

Main idea

What is the story mostly about?

Using knowledge and teamwork can help solve difficult problems.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Maggie, Travis, and Lucy**	Maggie and her teacher · Lucy and a scientist
Wanted	**to escape the room**	to find a hidden treasure · to build a science project
But	**the key was trapped inside ice**	the lights in the room went out · they lost the key on the floor
So	**they used what Lucy learned in science**	they broke the door open · they called a grown-up for help
Then	**the ice melted and they escaped**	they stayed in the room all day · the key broke into pieces

Model summary

Maggie, Travis, and Lucy wanted to get out of the room, but the key was stuck in ice. So Lucy used a science idea with a metal tray. Then the ice melted and they got out.

5 Seeking a Friend

Level 5 · Session 2 · Average · Sunny Meadow

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character?	**Mateo**	Santiago · Mateo's mom
2	What did Mateo want?	**A kitten**	A bicycle · A rabbit
3	What problem did Mateo face?	**Santiago needed extra care**	The shelter was closed · Santiago was missing
4	What did Mateo do?	**He thought about Santiago's needs**	He chose another cat · He left the shelter
5	What happened in the end?	**Mateo chose Santiago**	Mateo chose another pet · Mateo went home alone

Main idea

What is the story mostly about?

Kindness means caring for others even when they need extra help.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Mateo**	Santiago · Mateo's mom
Wanted	**a pet to bring home**	a new bicycle for his birthday · to work at the animal shelter
But	**Santiago was old and needed extra care**	the animal shelter was closed · his mom said no to any pet
So	**Mateo spent time with him and thought carefully**	Mateo picked a young kitten instead · Mateo went home to think alone
Then	**he chose Santiago as his friend**	he left Santiago at the shelter · he chose a rabbit as his pet

Model summary

Mateo wanted a pet cat, but Santiago was old and sick and needed more help. So Mateo spent time with him and thought hard. Then he picked Santiago as his new friend.

6 Shawn the Speedy Snail

Level 6 · Session 2 · Hard · Sunny Meadow

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Shawn**	The hungry snails · The strongest ditch
2	What did Shawn want?	**To get the food and water first**	To share with everyone · To leave the ditch
3	What problem did Shawn cause?	**He made the other snails hungry by being selfish**	He lost the food · He could not find water
4	What did Shawn realize?	**His actions had hurt his friends**	His friends were doing well · The ditch was empty
5	What happened in the end?	**Shawn helped his friends and changed his behavior**	Shawn stayed selfish · Shawn left forever

Main idea

What is the story mostly about?

People should use their talents to help others instead of thinking only about themselves.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Shawn**	The hungry snails · A big frog
Wanted	**to get all the food and water first**	to share his food with everyone · to find a new home in a pond
But	**his selfish actions made his friends hungry and upset**	a storm washed away all the food · he was too slow to reach the water
So	**he moved to another ditch and later helped his friends**	he hid the food from his friends · he stayed and ate even more food
Then	**he used his speed to help others instead of himself**	he kept all the food for himself · he never saw his friends again

Model summary

Shawn the snail wanted to get the food and drinks before anyone else, but the other snails got hungry and mad. So he left and came back to help. Then he used his speed for others.

7 Emma's Favorite Restaurant

Level 7 · Session 3 · Easy · River Bend

Page questions

Page	Question	Right answer	Other choices
1	Who is one of the main characters?	**Emma**	The chef · The waiter
2	What did Emma and Josh want?	**To eat at their favorite restaurant**	To stay home · To cook dinner
3	What problem did Emma and Josh face?	**They had to try a new restaurant they were unsure about**	Their favorite restaurant was closed · They forgot to eat dinner
4	What did Emma and Josh do?	**They listened to their parents and agreed to try it**	They refused to go · They ordered pizza
5	What happened in the end?	**They decided to try the new restaurant**	They stayed at home · They went to the Italian restaurant

Main idea

What is the story mostly about?
Trying new experiences can help people discover things they may enjoy.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Emma and Josh**	The waiter and the chef · Emma's father and mother
Wanted	**to eat at their favorite restaurant**	to cook dinner at home · to eat at a new Lebanese restaurant
But	**their father wanted to try a new restaurant**	their favorite restaurant was closed · they forgot to order their food
So	**they agreed to give the new restaurant a chance**	they stayed home and ordered pizza · they refused to go anywhere
Then	**they became willing to try something new**	they never ate out again · they went back to the Italian restaurant

Model summary

Emma and Josh wanted to keep eating at their favorite Italian place, but their dad picked a new Lebanese one. So they agreed to try it. Then they were willing to taste something new.

8 The Case of the Missing Lunch

Level 8 · Session 3 · Average · River Bend

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character?	**Timmy**	Sarah · Jake
2	What did Timmy want?	**To find his missing lunchbox**	To buy a new lunchbox · To go home
3	What problem did Timmy face?	**His lunchbox was missing**	He forgot his homework · He lost his backpack
4	What did Timmy and his friends do?	**They searched for clues and looked outside**	They stopped looking · They asked the teacher for a new lunchbox
5	What happened in the end?	**Timmy found his lunchbox with his friends' help**	The lunchbox stayed missing · Timmy bought a new lunchbox

Main idea

What is the story mostly about?
Working together and asking for help can solve problems more easily.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Timmy**	Jake · The teacher
Wanted	**to find his missing lunchbox**	to buy a new lunchbox · to play on the swings
But	**he could not find it by himself**	his friends would not talk to him · the playground was closed
So	**his friends helped him search**	he asked his mom for a new one · he went home early
Then	**they found the lunchbox near the swings**	they gave up looking · the lunchbox was never found

Model summary

Timmy wanted to get back his lost lunchbox, but he could not find it alone. So his friends helped him look for clues. Then they found it by the swings.

9 The Selfish Giant

Level 9 · Session 3 · Hard · River Bend

Page questions

Page	Question	Right answer	Other choices
1	Who is one of the main characters?	**The Giant**	The birds · The flowers
2	What did the children enjoy?	**Playing in the beautiful garden**	Climbing mountains · Visiting a castle
3	What problem began in the story?	**The Giant became angry when he saw the children**	The garden disappeared · The children broke the trees
4	What did the Giant do?	**He chased the children away and built a wall**	He invited the children to stay · He planted more trees
5	What happened in the end?	**The children could no longer play in the garden**	The children continued playing · The Giant shared the garden

Main idea

What is the story mostly about?
Being selfish can hurt others and take away their happiness.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**The Giant**	The singing birds · The children's teacher
Wanted	**to keep the garden only for himself**	to share the garden with everyone · to plant more fruit trees
But	**the children loved playing there**	the flowers in the garden died · the birds stopped singing
So	**he chased them away and built a wall**	he invited them to stay and play · he moved far away again

Part	Right answer	Wrong race choices
Then	**the children lost their favorite place to play**	the children played there every day · the Giant shared his garden

Model summary

The Giant wanted his garden all to himself, but the kids loved to play there. So he got angry, sent them away and built a wall. Then the sad children lost their happy place.

10 A Visit to the Zoo

Level 10 · Session 4 · Easy · River Bend

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character?	**Omar**	The lion · The giraffe
2	What did Omar's family want to do?	**Visit the zoo and see animals**	Go shopping · Watch a movie
3	What animals did Omar enjoy watching?	**Giraffes and monkeys**	Sharks and whales · Rabbits and cats
4	What activity did Omar's sister enjoy?	**Feeding the ducks**	Riding a train · Taking pictures
5	How did Omar feel at the end?	**Excited and happy**	Sad · Angry

Main idea

What is the story mostly about?

Spending time with family and exploring new places can create happy memories.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Omar and his family**	Omar's teacher and class · The zookeeper and the lions
Wanted	**to enjoy a day at the zoo**	to go shopping at the mall · to watch a movie at home
But	**they had many animals to explore**	the zoo was closed that day · it rained all day long
So	**they visited different animal exhibits**	they went straight back home · they only looked at the lions
Then	**they had a fun day and made happy memories**	they felt tired and bored · they got lost at the zoo

Model summary

Omar and his family wanted a fun day at the zoo, but there were so many animals to explore. So they went to see lions, giraffes and ducks. Then they made happy memories together.

11 Be Careful What You Wish For

Level 11 · Session 4 · Average · River Bend

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Rylee**	Uncle Aaron · Belinda
2	What did Rylee want?	**Her classmates to like the necklace**	A new necklace · To go home
3	What problem did Rylee notice?	**Her wishes seemed to cause unexpected problems**	She lost her test · The necklace broke
4	What did Rylee do to solve her problem?	**She asked the museum for information about it**	She hid the necklace · She gave it to a friend
5	What happened in the end?	**Rylee decided she no longer wanted the necklace**	Rylee kept using the necklace · Rylee wished for more things

Main idea

What is the story mostly about?

People should think carefully about what they wish for because actions can have unexpected consequences.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Rylee**	Uncle Aaron · Rylee's teacher
Wanted	**good luck and for her wishes to come true**	to give the necklace to a friend · to visit a museum with her class
But	**her wishes seemed to have unexpected consequences**	the necklace broke into pieces · her uncle took the necklace back
So	**she investigated the necklace and learned more about it**	she made even more wishes · she hid the necklace under her bed
Then	**she decided to get rid of the necklace**	she kept wishing for more things · she wore the necklace every day

Model summary

Rylee wanted her wishes to come true, but each wish caused trouble for others. So she went to a museum to learn about the necklace. Then she decided to stop keeping it.

12 AgitAgueda

Level 12 · Session 4 · Hard · River Bend

Page questions

Page	Question	Right answer	Other choices
1	Who are the main subjects in the story?	**Festival visitors**	Farmers · Sailors
2	What did the visitors want?	**To enjoy Portugal's festivals and culture**	To buy a new house · To build umbrellas
3	What challenge did the visitors face?	**They had many festivals and attractions to explore.**	They had nothing to see. · They could not enter Portugal.
4	How did the visitors experience Portugal's culture?	**By attending festivals and visiting famous attractions**	By staying inside their hotel · By avoiding local traditions
5	What happened in the end?	**The visitors experienced Portugal's festivals and culture.**	The visitors left without learning anything. · The festival was canceled.

Main idea

What is the story mostly about?

Portugal's festivals allow visitors to experience the country's vibrant culture, traditions, and famous attractions.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**People who visit AgitAgueda and other festivals in Portugal**	Farmers who grow food in the fields of Portugal · Sailors who travel across the sea from Portugal
Wanted	**to enjoy music, art, food, and Portuguese culture**	to buy a new house in a quiet Portuguese town · to build umbrellas and sell them at the market
But	**there are many different festivals and attractions to explore**	the festival was canceled because of bad weather · they could not find any place to stay in Portugal
So	**they visited AgitAgueda and experienced its unique events and landmarks**	they stayed inside their hotel for the whole month · they went home before the festival started
Then	**they discovered Portugal's rich traditions and festival culture**	they left without learning anything about Portugal · they decided never to visit another festival

Model summary

Visitors wanted to enjoy music, art and food in Portugal, but there were so many festivals to explore. So they went to AgitAgueda and its landmarks. Then they discovered the country's rich traditions.

13 The Crowded House: A Folktale

Level 13 · Session 5 · Easy · Story Town

Page questions

Page	Question	Right answer	Other choices
1	Who are the main characters in the story?	**The Rubin family**	Reb Solman only · The village animals
2	What did the Rubin family want?	**More space in their home**	A bigger house · More visitors
3	What problem did the family face?	**Their house felt too crowded**	They lost their animals · They moved away
4	What did the family do to solve the problem?	**They followed Reb Solman’s advice**	They built a new house · They sold their animals
5	What happened in the end?	**The house felt spacious and the family was grateful**	The family moved away · The animals stayed inside

Main idea

What is the story mostly about?

People should appreciate what they have because things may be better than they seem.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**The Rubin family**	Reb Solman, the wise man · The relatives and the animals

Part	Right answer	Wrong race choices
Wanted	**a less crowded house**	a bigger farm with more animals · more visitors in their house
But	**their house felt too small and cramped**	their animals ran away from the farm · Reb Solman would not help them at all
So	**they followed Reb Solman's unusual advice**	they built a new room on their house · they moved to a bigger village far away
Then	**they realized their house was not crowded after all**	they kept all the animals inside forever · they moved into Reb Solman's big house

Model summary

The Rubin family wanted more room in their small home, but it felt too crowded. So they listened to Reb Solman and brought in relatives and animals. Then the house felt big when everyone left.

14 Owen and Mzee

Level 14 · Session 5 · Average · Story Town

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Owen**	Mzee · The zookeeper
2	What did Owen want?	**Comfort and friendship**	A new cage · More food
3	What problem did Owen face?	**Mzee did not want to be his friend at first**	He could not swim · He wanted to leave the park
4	What did Owen do?	**He kept trying to be near Mzee**	He ignored Mzee · He found another friend
5	What happened in the end?	**Owen and Mzee became close friends**	Owen left the park · Mzee moved away

Main idea

What is the story mostly about?

Friendship can grow between individuals who are very different from one another.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Owen**	Mzee · The scientists
Wanted	**comfort and friendship**	food and a warm bed · to swim back to the sea
But	**he was alone and Mzee avoided him**	he could not find any food · the park was too cold for him
So	**Owen kept staying close to Mzee**	Owen swam back to the sea · Owen found a new hippo friend
Then	**they became close friends**	Owen left the wildlife park · Mzee moved far away

Model summary

Owen the baby hippo wanted a friend, but he was alone and the old tortoise Mzee stayed away from him. So Owen kept following him. Then they became best buddies.

15 In Grandfather's Day

Level 15 · Session 5 · Hard · Story Town

Page questions

Page	Question	Right answer	Other choices
1	Who is sharing stories about the past?	**Grandfather**	Sharr · Kaze
2	What did Grandfather want to do?	**Share memories about the past**	Teach them how to drive · Show them a movie
3	What challenge did the grandchildren have?	**They found it hard to imagine life in the past**	They forgot the stories · They did not like technology
4	How did Grandfather help them understand?	**He shared stories and memories**	He gave them homework · He bought them books
5	What happened in the end?	**The grandchildren appreciated learning about the past**	They ignored Grandfather · They left early

Main idea

What is the story mostly about?
Learning about the past helps us appreciate how life has changed over time.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Grandfather**	Sharr and Kaze · Grandfather's mother
Wanted	**to share memories of life in the past**	to buy new Mindcaps for his grandchildren · to teach the children how to type fast
But	**his grandchildren found it hard to imagine**	he forgot all of his old stories · his grandchildren did not visit him
So	**he showed them pictures and told stories**	he gave them a lot of homework · he took them for a ride in a car
Then	**they appreciated how different life used to be**	they forgot what Grandfather said · they wanted to go home early

Model summary

Grandfather wanted to talk about his childhood, but Sharr and Kaze could not imagine a world without Mindcaps. So he showed them photos and told stories. Then they were amazed.

16 You Can't Always Tell

Level 16 · Session 6 · Easy · Story Town

Page questions

Page	Question	Right answer	Other choices
1	Who are the main characters in the story?	**A father and his son**	The wagon driver · The innkeeper
2	What did the father and son want?	**To find safety after the flood**	To buy a wagon · To move to another country
3	What problem did they face?	**Many things seemed to go wrong for them**	They found too much treasure · They lost the treasure chest
4	What did the father do when problems happened?	**He stayed hopeful and kept going**	He gave up · He blamed others
5	What happened in the end?	**They found treasure and realized their luck was good**	They remained poor · They lost everything again

Main idea

What is the story mostly about?
Things that seem like bad luck can sometimes lead to good outcomes.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**A father and his son**	The wagon driver and his wife · The innkeeper and his family
Wanted	**to find safety and rebuild their lives after the flood**	to buy a big wagon and travel to the city · to find a treasure chest hidden in the river
But	**many things kept going wrong for them**	the father did not want to leave · they had too much money to carry
So	**they continued moving forward and stayed hopeful**	they gave up and stayed in the stable · they blamed the innkeeper for everything
Then	**they discovered that their bad luck led to good outcomes**	they lost everything again in another flood · they never went back to their old home

Model summary

A father and son wanted a safe new home after a flood, but they kept having bad luck. So they stayed hopeful and kept going. Then they found treasure when they got home.

17 Shuffle to Buffalo

Level 17 · Session 6 · Average · Story Town

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Amelia**	Mom · The audience
2	What did Amelia want?	**To perform her solo successfully**	To skip the recital · To become a singer
3	What problem did Amelia face?	**She was nervous about making mistakes**	She forgot her costume · She missed the recital
4	What did Amelia do to solve the problem?	**She practiced and continued dancing confidently**	She stopped dancing · She left the stage
5	What happened in the end?	**Amelia performed successfully and felt proud of herself**	Amelia quit dancing · The recital was canceled

Main idea

What is the story mostly about?
Practice and confidence helped Amelia overcome her fear and succeed in her dance recital.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Amelia**	Amelia's mom · The audience
Wanted	**to perform her tap dance solo successfully**	to sing a song at the school concert · to skip the recital and stay home

Part	Right answer	Wrong race choices
But	**she was nervous about making mistakes during the recital**	she lost her tap shoes before the recital · her mom could not come to watch her dance
So	**she practiced everywhere and stayed confident during her performance**	she stopped dancing and left the stage · she asked her teacher to dance for her
Then	**she finished the recital successfully and felt proud of herself**	she quit dancing and never went back · the recital was canceled because of rain

Model summary

Amelia wanted to dance her tap solo well, but she was scared of messing up in front of everyone. So she practiced a lot and stayed brave. Then the crowd clapped and she felt proud.

18 In Grandfather's Day

Level 18 · Session 6 · Hard · Story Town

Page questions

Page	Question	Right answer	Other choices
1	Who are the main characters learning about the past?	**Sharr and Kaze**	Owen and Mzee · Papa and Mama
2	What did Sharr and Kaze want to learn?	**What life was like in the past**	How to build a robot · How to cook stew
3	What was different about life long ago?	**People typed on keyboards and drove cars**	People used thought commands · People lived on the moon
4	What memory did Grandfather share?	**Cooking stew with his mother**	Flying a spaceship · Riding a Plugger
5	What happened in the end?	**The children learned to appreciate life in the past**	The children disliked the stories · Grandfather forgot his memories

Main idea

What is the story mostly about?

Learning about the past can help us appreciate how people lived and the experiences they enjoyed.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Sharr and Kaze**	Grandfather and his mother · Owen and Mzee
Wanted	**to learn about life in Grandfather's day**	to learn how to build a new robot · to buy Grandfather a car for his birthday
But	**life in the past was very different from their futuristic world**	Grandfather did not remember anything · they had no time to visit Grandfather
So	**Grandfather shared stories and memories from his childhood**	Grandfather sent them home to read a book · Grandfather bought them new Pluggers
Then	**the children gained a new appreciation for life in the past**	the children forgot all of the stories · the children wanted Mindcaps for everyone

Model summary

Sharr and Kaze wanted to know about Grandfather's childhood, but their future world with Mindcaps was so different. So he told them stories about cars and stew. Then they saw the past was special.

19 The Boy With the Ball

Level 19 · Session 7 · Easy · Palm Beach

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Hector**	The teacher · Josh
2	What did Hector want?	**To make friends**	To move away · To buy a new ball
3	What problem did Hector face?	**He did not know anyone at school**	He lost his ball · He missed the bus
4	What did Hector do to help solve his problem?	**He showed his ball skills**	He hid from his classmates · He went home
5	What happened in the end?	**Hector made new friends through his talent**	Hector remained lonely · Hector stopped playing with the ball

Main idea

What is the story mostly about?

Sharing your talents can help you build friendships and confidence.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Hector**	Josh · The teacher
Wanted	**to make friends at his new school**	to move back to his old town · to win a ball game at school
But	**he was new and did not know anyone**	he lost his favorite ball · he missed the first day of school
So	**he showed his ball tricks during recess**	he stayed in the classroom during recess · he gave his ball away to a classmate
Then	**his classmates admired his skills and became his friends**	his classmates laughed and walked away from him · he decided to stop playing with the ball forever

Model summary

Hector wanted to find buddies at his new school, but he was nervous since nobody knew him. So he did cool ball tricks at recess. Then the kids liked him and became his friends.

20 Sploosh!

Level 20 · Session 7 · Average · Palm Beach

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Chef Pierre**	Slice · Queen Penelope
2	What did Chef Pierre want?	**To become the Queen's Chef Royale**	To catch fish · To grow oranges
3	What problem did Chef Pierre face?	**The frog kept escaping**	The queen canceled the contest · His kitchen was closed
4	What did Chef Pierre do to solve the problem?	**He tried to catch the frog himself**	He asked the queen for help · He cooked another meal
5	What happened in the end?	**Slice and Dice became the Chefs Royales, and Pierre became the Frog Keeper**	Chef Pierre became the Queen's Chef Royale · The frog became the queen's pet

Main idea

What is the story mostly about?
Being overconfident does not always lead to success, while teamwork and humility can bring better results.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Chef Pierre**	Queen Penelope · Slice and Dice
Wanted	**to become the Queen's Chef Royale by cooking his famous frog legs**	to become the Frog Keeper by taking care of the frogs in the pond · to open his own restaurant by selling marmalade sandwiches
But	**he and his assistants could not catch the frog**	the queen did not like his messy kitchen · his assistants ate all of the frog legs
So	**Chef Pierre tried to catch the frog himself**	Chef Pierre asked the queen for some help · Chef Pierre cooked a different meal instead
Then	**he failed, and Slice and Dice became the Queen's Chefs Royales while Pierre became the Frog Keeper**	he won, and Slice and Dice became the Frog Keepers while Pierre became the Queen's Chef Royale · he gave up, and the queen made the frog her new pet while Pierre went back home

Model summary

Chef Pierre hoped to cook frog legs so the queen would make him her top chef, but nobody could catch the frog. So he tried it himself and fell in. Then Slice and Dice got the job.

21 First Day at the Factory

Level 21 · Session 7 · Hard · Palm Beach

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Stanley Marks**	Mr. Hubbard · Henry Ford
2	What did Stanley want?	**To succeed at his job and support his family**	To quit immediately · To become the factory manager
3	What problem did Stanley face?	**The work was difficult and exhausting.**	The factory had no workers. · He forgot where the factory was.

Page	Question	Right answer	Other choices
4	How did Stanley respond to the challenge?	**He learned his job and worked hard.**	He refused to work. · He left during training.
5	What happened in the end?	**Stanley continued working to provide for his family.**	Stanley quit his job. · Stanley became the owner of Ford.

Main idea

What is the story mostly about?

Hard work can be challenging, but perseverance and responsibility help people provide a better future for their families.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Stanley Marks**	Mr. Hubbard · Henry Ford
Wanted	**to succeed at his new factory job and support his family**	to become the boss of the whole factory and its workers · to buy a Model T car on his very first day of work
But	**the work was difficult, repetitive, and physically demanding**	the factory was closed, empty, and far away from home · the boss was angry, loud, and unfair to every worker
So	**he learned his task and worked hard on the assembly line**	he quit his task and went home before the lunch break · he watched the other workers and rested near the line
Then	**he accepted the challenges because the job would provide a better future for his family**	he quit the job because the work was too hard for him to finish each day · he became the owner of Ford and gave his family a big new house

Model summary

Stanley Marks wanted to succeed at his job to support his family, but the factory work was hard and tiring. So he learned his task and worked hard. Then he stayed for a better future.

22 An Ice Idea

Level 22 · Session 8 · Easy · Palm Beach

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Doris**	Charlie · Papa
2	What did the family want?	**To keep their food cold and fresh**	To buy a car · To move away
3	What problem did the family face?	**Their icebox could not keep food cold enough**	Their refrigerator broke · They lost electricity
4	What did the family do to solve the problem?	**Purchased a new refrigerator**	Bought more ice · Stopped buying food
5	What happened in the end?	**The family's food stayed fresh, but Doris worried about Charlie**	The refrigerator did not work · Charlie bought the refrigerator

Main idea

What is the story mostly about?

New inventions can make life easier, but they can also change people's jobs.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Doris and her family**	Charlie and his friends · The neighbors next door
Wanted	**to keep their food fresh and cold**	to buy a shiny new family car · to sell ice to all their neighbors
But	**the icebox could not keep enough ice during the hot summer**	the store ran out of food during the long, cold winter · the new refrigerator broke down on the very first day
So	**Mama bought a new electric refrigerator**	Papa bought a much bigger block of ice · Mama asked Charlie to come every day
Then	**their food stayed fresh, though Doris worried about the ice man's future**	their food spoiled again, and Doris was angry at the ice man for it · their refrigerator broke, and Doris had to buy more ice from Charlie

Model summary

Doris and her family needed to keep their food cool, but the ice kept melting in the hot summer. So Mama bought an electric fridge. Then the food stayed fresh but Doris felt sorry for Charlie.

23 How to Skateboard

Level 23 · Session 8 · Average · Palm Beach

Page questions

Page	Question	Right answer	Other choices
1	Who is the passage about?	**A beginner skateboarder**	A bicycle rider · A race car driver
2	What does a beginner need before skateboarding?	**A skateboard and protective gear**	A motorcycle · A surfboard
3	What challenge does the beginner face?	**Learning balance and control**	Finding food · Building a skateboard
4	What does the beginner do to improve?	**Practices basic skateboarding skills**	Watches television · Buys a new skateboard every day
5	What happens in the end?	**The beginner becomes ready for more advanced tricks**	The beginner quits skateboarding · The skateboard breaks

Main idea

What is the story mostly about?

Practice, patience, and safety help beginners learn how to skateboard.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**A beginner skateboarder**	A skilled bicycle rider · A race car driver
Wanted	**to learn how to skateboard**	to learn how to swim · to build a new skateboard
But	**skateboarding requires balance, practice, and safety skills**	skateboarding requires money, tickets, and a big team · skateboards are very hard to find in most stores
So	**the beginner learned the basic movements step by step**	the beginner tried a big trick on the very first day · the beginner watched videos and never practiced at all

Part	Right answer	Wrong race choices
Then	**they became ready to learn more advanced tricks**	they gave up and put the skateboard away · they broke the skateboard and stopped riding

Model summary

A beginner wanted to learn to ride a skateboard, but it takes balance and safe habits. So the beginner practiced pushing and braking. Then they felt confident and ready for tricks.

24 Anansi and the Cook Pots

Level 24 · Session 8 · Hard · Palm Beach

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Anansi**	Rabbit · Fox
2	What did Anansi want?	**To eat food from all his friends**	To cook food · To build a house
3	What problem did Anansi face?	**All the cook pots were ready at once**	He forgot where the food was · His friends left the village
4	What happened when the friends pulled the webs?	**His legs were pulled in different directions**	Anansi got all the food · The webs broke
5	What happened in the end?	**Anansi learned a lesson about greed**	Anansi became rich · Anansi ate all the food

Main idea

What is the story mostly about?

Greed can cause problems and lead to unpleasant consequences.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Anansi**	Rabbit · Fox
Wanted	**to eat food from all of his friends without cooking**	to cook a big meal for all of his friends in the village · to build a new house right next to the river
But	**all of the cook pots became ready at the same time**	all of his friends forgot to cook any food that day · all of the webs broke before the food was ready
So	**every web attached to his legs was pulled at once**	every friend brought the food straight to his house · every cook pot spilled into the river at once
Then	**his legs stretched out, and he learned a lesson about greed**	his legs got shorter, and he ate all of the food himself · his friends got angry, and he moved to a new village

Model summary

Anansi the spider wanted to eat his friends' food without cooking, but every pot was ready at once. So all the webs on his legs got pulled. Then his legs grew long and he learned about greed.

25 A Pool Fit for a Hedgehog

Level 25 · Session 9 · Easy · Summit Peak

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character?	**Cora**	Heather · The pool company
2	What did Cora want?	**A pool to stay cool**	A larger house · More food
3	What problem did Cora face?	**She did not know which company to choose**	She did not like swimming · She lost her money
4	What did Cora do to solve her problem?	**She researched and compared companies**	She guessed · She asked no questions
5	What happened in the end?	**Cora bought a pool and enjoyed it**	Cora changed her mind · Cora sold the pool

Main idea

What is the story mostly about?
Researching before making a decision can help people choose wisely.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Cora**	Heather · The pool company
Wanted	**a pool to stay cool during the summer**	a bigger house to live in · a new friend to play with
But	**she needed to decide which company to buy from**	she did not know how to swim at all · she lost all of her money at the store
So	**she researched different pool companies and reviews**	she asked Heather to give her the pool · she guessed and picked the first one
Then	**she bought a pool and enjoyed using it**	she changed her mind and stayed home · she sold the pool to Heather

Model summary

Cora the hedgehog wanted her own pool to keep cool, but she had to choose which company to buy from. So she looked at reviews and prices. Then she bought one and enjoyed it.

26 First Fast

Level 26 · Session 9 · Average · Summit Peak

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Asma**	Her mother · Her teacher
2	What did Asma want?	**To complete her first Ramadan fast**	To eat more snacks · To stay home from school
3	What problem did Asma face?	**She became hungry while fasting**	She forgot to eat breakfast · She lost her gifts
4	What did Asma do to solve the problem?	**She continued fasting and followed the Ramadan practices**	She stopped fasting · She skipped school

Page	Question	Right answer	Other choices
5	What happened in the end?	**Asma celebrated Eid al-Fitr and felt proud of herself**	Asma gave up fasting · Asma stayed home alone

Main idea

What is the story mostly about?

Perseverance, faith, and self-discipline helped Asma successfully complete her first Ramadan fast and celebrate her achievement.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Asma**	Asma's mother · Asma's teacher
Wanted	**to complete her first Ramadan fast**	to eat more snacks during the day · to get new clothes for school
But	**she became hungry during the day and missed eating snacks**	she forgot to wake up before sunrise every morning · she lost the gifts her family gave her for Eid
So	**she continued fasting and followed the Ramadan practices**	she stopped fasting and ate snacks with her friends · she skipped school and stayed in bed all day
Then	**she celebrated Eid al-Fitr and felt proud of completing her first fast**	she gave up fasting and felt sad about it · she stayed home alone and missed the feast

Model summary

Asma wanted to finish her first Ramadan fast, but she got very hungry and missed her snacks. So she kept going with her family's traditions. Then she celebrated Eid and felt proud.

27 Baba Yaga, the Girl, and the Hedgehog

Level 27 · Session 9 · Hard · Summit Peak

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Marusia**	Baba Yaga · Dmitri
2	What did Marusia want?	**To escape and return home**	To become a witch · To stay with Baba Yaga
3	What problem did Marusia face?	**She was trapped by Baba Yaga.**	She lost her family. · She forgot the way to the market.
4	How did Marusia solve the problem?	**She worked with the hedgehog to find the magical flower.**	She fought Baba Yaga. · She hid in the forest.
5	What happened in the end?	**Marusia and Dmitri were freed and safely returned home.**	Marusia remained trapped forever. · Baba Yaga became queen.

Main idea

What is the story mostly about?

Wisdom, cooperation, and courage helped Marusia overcome danger and regain her freedom.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Marusia**	Baba Yaga · Dmitri the prince
Wanted	**to escape Baba Yaga and return home**	to learn magic from Baba Yaga · to find a pet hedgehog in the forest
But	**Baba Yaga captured her and magically locked the gate**	she got lost on the way to the market · the hedgehog ran away from her
So	**Marusia worked with the talking hedgehog to find the magical flower**	Marusia fought Baba Yaga with a broom · Marusia hid in the forest until dark
Then	**Baba Yaga released them, and Marusia and Dmitri returned home safely**	Marusia stayed with Baba Yaga forever · Baba Yaga became the queen of the land

Model summary

Marusia wanted to get away from Baba Yaga and go home, but the witch locked her in. So she teamed up with a talking hedgehog to find the magic flower. Then they were set free.

28 Why Does the Ocean Have Waves?

Level 28 · Session 10 · Easy · Summit Peak

Page questions

Page	Question	Right answer	Other choices
1	What is the topic of the passage?	**Ocean waves**	Fish · Boats
2	What is the most common cause of waves?	**Wind**	Fish · Ships
3	What problem can severe weather cause?	**Storm surges**	Rainbows · Icebergs
4	What can create a tsunami?	**Underwater disturbances like earthquakes**	Wind only · Clouds
5	What happens in the end of the explanation?	**Different forces work together to create waves and tides**	The ocean stops moving · Tsunamis become tides

Main idea

What is the story mostly about?

Ocean waves are created by different natural forces such as wind, storms, earthquakes, and gravity.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Ocean waves**	Ocean fish · Big ships
Wanted	**to move energy across the water**	to carry boats to the beach · to make the water warm
But	**different natural forces create different kinds of waves**	the ocean is too deep for waves to form · fish swim fast and stop the waves
So	**waves form through wind, storms, earthquakes, and gravity**	waves form only when ships sail by · waves form when clouds touch the sea
Then	**the ocean remains in constant motion**	the ocean becomes calm and still · the waves freeze into ice

Model summary

Ocean waves move energy through the sea, but many kinds of natural forces make them. So wind, storms, earthquakes, and the moon's gravity all push the water. Then the ocean keeps moving all the time.

29 Clara Barton: Civil War Hero

Level 29 · Session 10 · Average · Summit Peak

Page questions

Page	Question	Right answer	Other choices
1	Who is the main character in the story?	**Clara Barton**	A soldier · A doctor
2	What did Clara Barton want?	**To help people in need**	To become rich · To become famous
3	What problem did Clara face?	**Soldiers needed help during the war**	There were no schools · She lost her job
4	What did Clara do to help solve the problem?	**She gathered supplies and cared for wounded soldiers**	She ignored the soldiers · She stayed at home
5	What happened in the end?	**Clara founded the American Red Cross and became a hero**	Clara stopped helping others · Clara became a soldier

Main idea

What is the story mostly about?

Clara Barton dedicated her life to helping others and founded the American Red Cross.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Clara Barton**	A wounded soldier · A Civil War doctor
Wanted	**to help people in need**	to become rich and famous · to become a soldier in the war
But	**many soldiers and families suffered during and after the Civil War**	she lost her job as a teacher · there were no schools for the children
So	**she gathered supplies, cared for soldiers, and searched for missing people**	she stayed home and wrote letters to friends · she ignored the soldiers and went away
Then	**she founded the American Red Cross and became a respected hero**	she stopped helping and became a soldier · she went back to teaching and forgot the war

Model summary

Clara Barton wanted to help people, but many soldiers were hurt in the Civil War. So she brought supplies and cared for them. Then she started the American Red Cross.

30 We Also Serve

Level 30 · Session 10 · Hard · Summit Peak

Page questions

Page	Question	Right answer	Other choices
1	Who are the main subjects of the story?	**Brave service animals**	Farmers · Scientists
2	What did Maria Dickin want?	**To honor brave animals**	To train animals · To start a zoo
3	What challenge did the animals face?	**They worked in dangerous situations**	They had no homes · They could not find food
4	How did the animals help people?	**By bravely carrying out important tasks**	By performing tricks · By guarding farms
5	What happened in the end?	**The animals were honored with the Dickin Medal**	The animals were forgotten · The medal was removed

Main idea

What is the story mostly about?

Animals have shown great courage and loyalty in helping people and deserve recognition for their service.

Story parts (race and story order)

Part	Right answer	Wrong race choices
Somebody	**Brave animals that served people**	Farmers who raised animals · Scientists who studied pigeons
Wanted	**to help and protect others**	to win prizes and shiny medals · to find food and new homes
But	**they faced dangerous situations during conflicts and emergencies**	they had no homes and no food to eat · they were too small to do any work
So	**they worked loyally and courageously to save lives**	they did tricks to make people laugh · they ran away and hid from danger
Then	**they were honored with the Dickin Medal for their service**	they were forgotten after the war ended · the medal was taken away from them

Model summary

Brave animals like pigeons and guide dogs wanted to protect people, but they worked in danger during wars. So they carried messages and led their owners out. Then they won the Dickin Medal.